

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7852
Nombre de lignes df clean	7852
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5496
Taille dataset test	2355
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

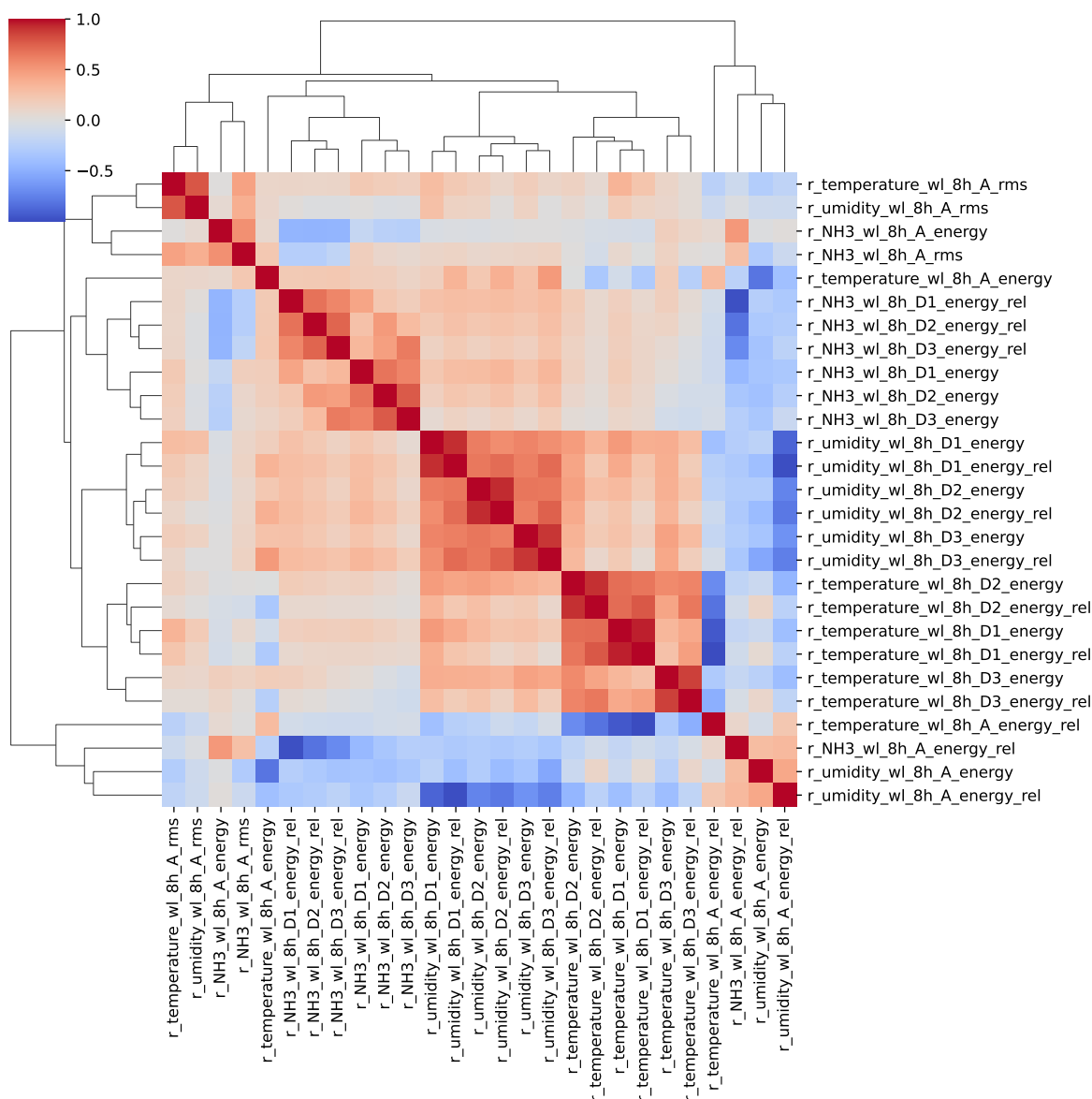
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_wl_8h_D1_energy ; **cluster 7:** r_NH3_wl_8h_D2_energy ; **cluster 8:** r_NH3_wl_8h_D3_energy ; **cluster 9:** r_NH3_wl_8h_A_energy ; **cluster 10:** r_NH3_wl_8h_A_rms ; **cluster 11:** r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D1_energy_rel ; **cluster 12:** r_NH3_wl_8h_D3_energy_rel ; **cluster 13:** r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D1_energy_rel ; **cluster 14:** r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_D3_energy_rel ; **cluster 15:** r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_energy ; **cluster 16:** r_temperature_wl_8h_A_rms ; **cluster 17:** r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D1_energy ; **cluster 18:** r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D2_energy_rel ; **cluster 19:** r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_D3_energy ; **cluster 20:** r_umidity_wl_8h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3-en-

ergy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms

Len(Features_X): 20

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

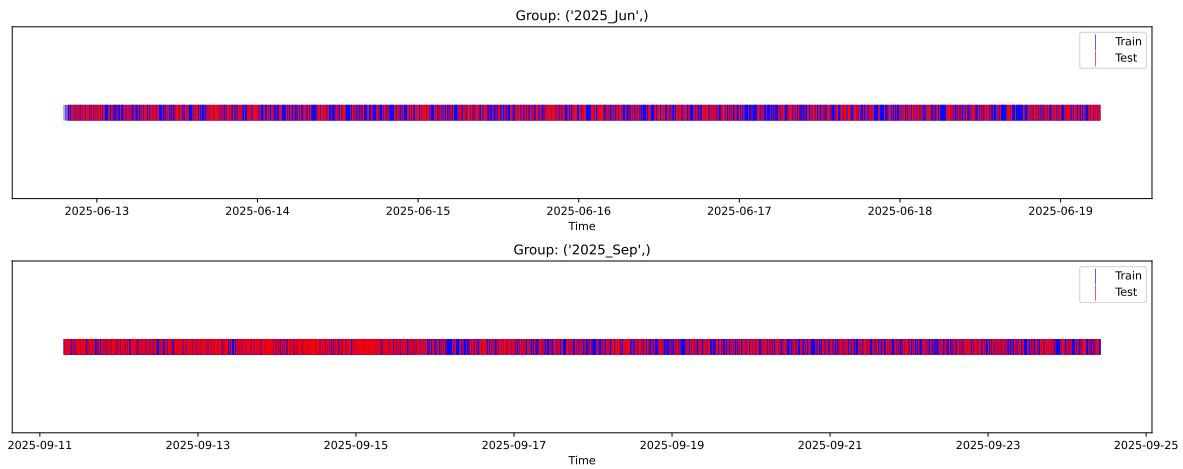
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5496 samples Test: 2356 samples



3.3.2 Internal CV Folds

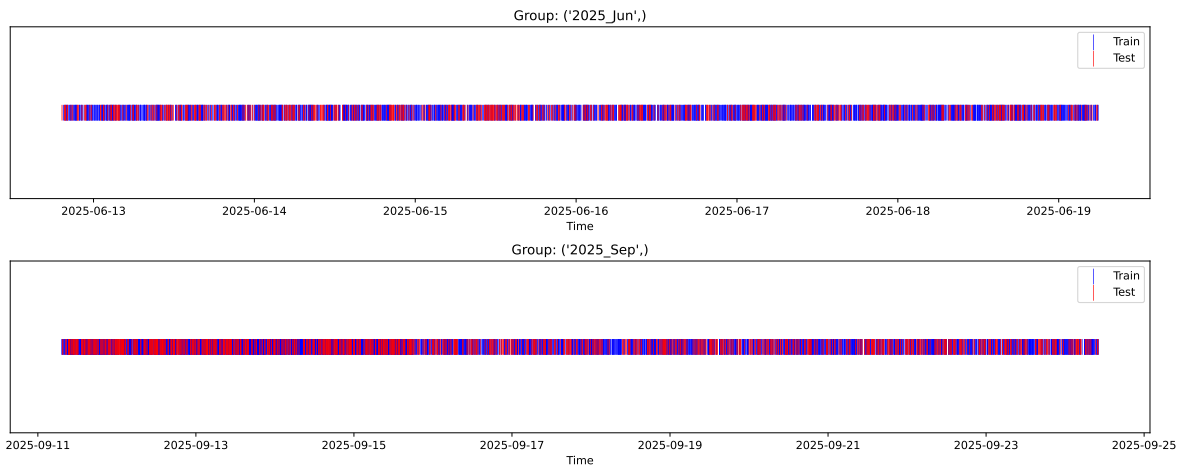
Fold 1/5

Train: 3847 samples Test: 1649 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3847 samples Test: 1649 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3847 samples Test: 1649 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3847 samples Test: 1649 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3847 samples Test: 1649 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.352, std=0.802, min=0.959, max=13.845
 y_{test} : mean=2.361, std=0.799, min=1.068, max=8.190

Fold 0 — y_{test} : mean=2.33, std=0.77, y_{train} : mean=2.36, std=0.82

Fold 1 — y_{test} : mean=2.39, std=0.77, y_{train} : mean=2.33, std=0.81

Fold 2 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.35, std=0.80

Fold 3 — y_test: mean=2.34, std=0.81, y_train: mean=2.36, std=0.80

Fold 4 — y_test: mean=2.34, std=0.79, y_train: mean=2.36, std=0.81

train scores CV: [0.53296676 0.52726644 0.53481009 0.53401602 0.52969049]

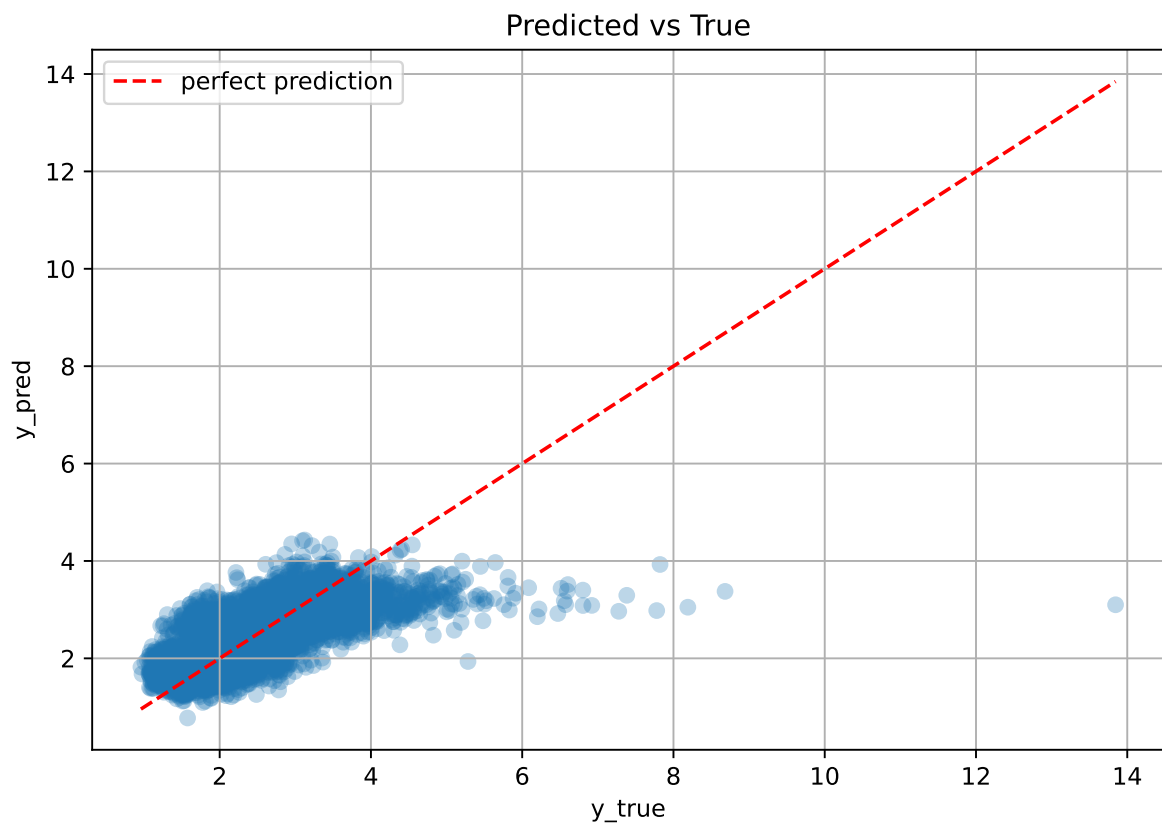
test scores CV: [0.54503299 0.55751411 0.53998231 0.54033457 0.55331776]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5317	0.5472	0.007	0.5285	0.5302	-0.017	0.9689	0.9717

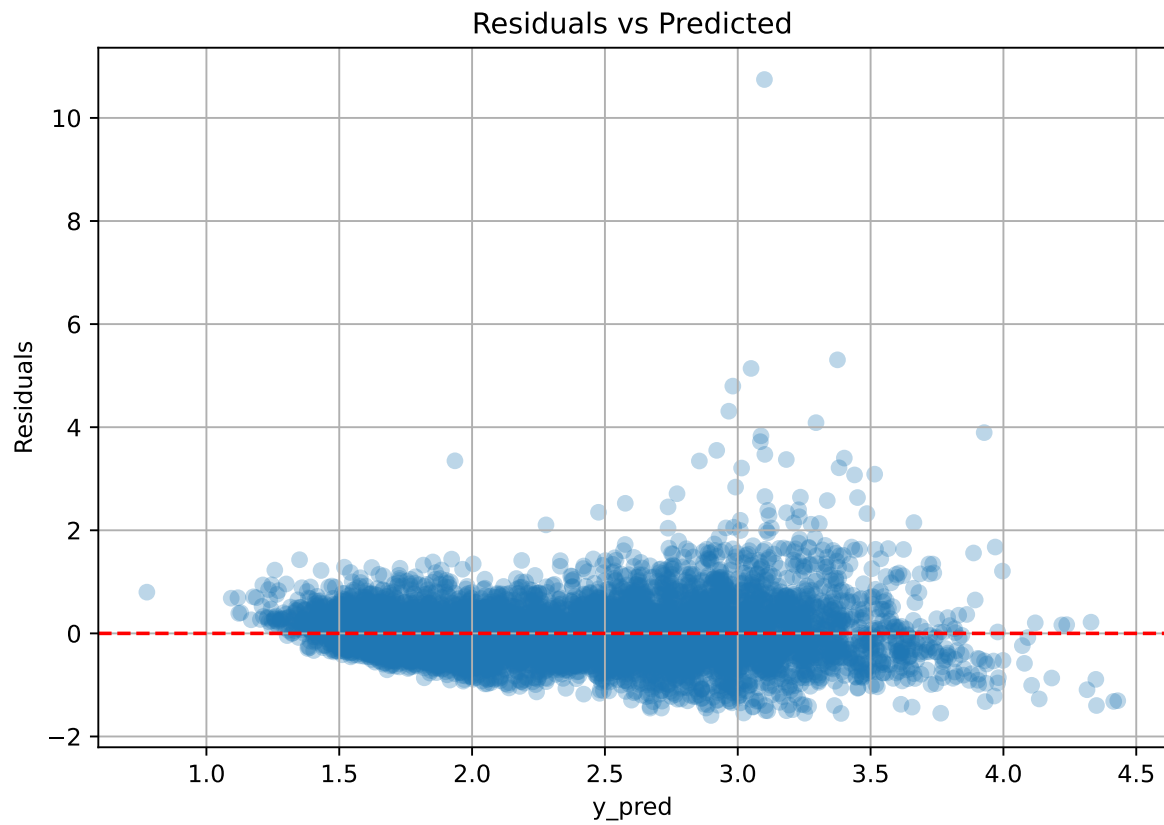
Coefficients:

coef_r_CO2 = -0.0499, coef_r_NH3 = 0.1675, coef_r_NH3_wl_8h_A_energy = -0.2100, coef_r_NH3_wl_8h_A_energy_rel = 0.0286, coef_r_NH3_wl_8h_A_rms = -0.0013, coef_r_NH3_wl_8h_D1_energy = 0.0642, coef_r_NH3_wl_8h_D2_energy = 0.1034, coef_r_NH3_wl_8h_D3_energy = -0.0353, coef_r_NH3_wl_8h_D3_energy_rel = -0.0589, coef_r_THI = 1.0194, coef_r_temperature = -1.1487, coef_r_temperature_wl_8h_A_energy_rel = -0.1895, coef_r_temperature_wl_8h_A_rms = -0.0479, coef_r_temperature_wl_8h_D3_energy = -0.1217, coef_r_umidity = -0.8124, coef_r_umidity_wl_8h_A_energy = 0.2122, coef_r_umidity_wl_8h_A_energy_rel = 0.0688, coef_r_umidity_wl_8h_A_rms = -0.0825, coef_r_umidity_wl_8h_D2_energy = -0.0159, coef_r_umidity_wl_8h_D3_energy_rel = 0.0865, intercept = 2.3524,

3.4.2 Scatter



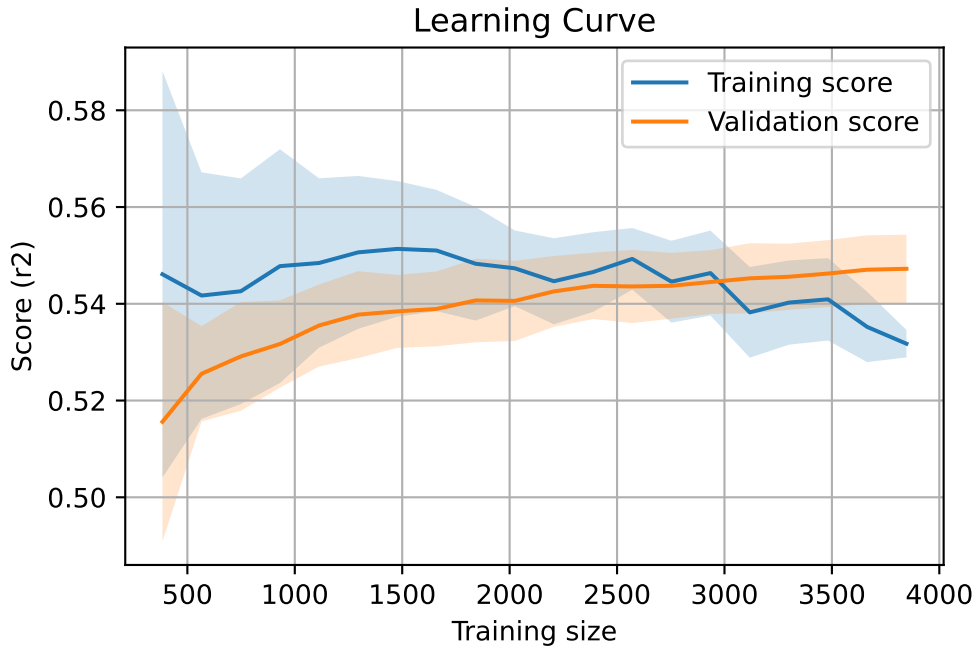
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7852, `Len(training)` : 5497,

`Len(training_real)` : 3848, `Len(test_of_training)` : 1649, `Len(test)` : 2355



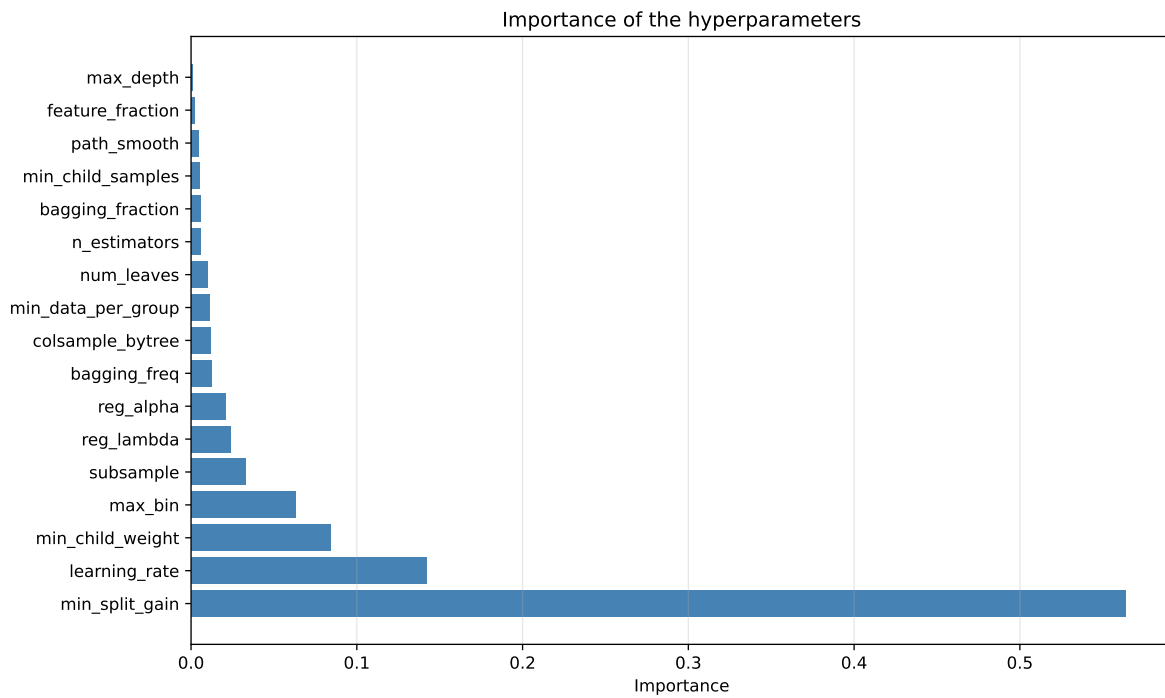
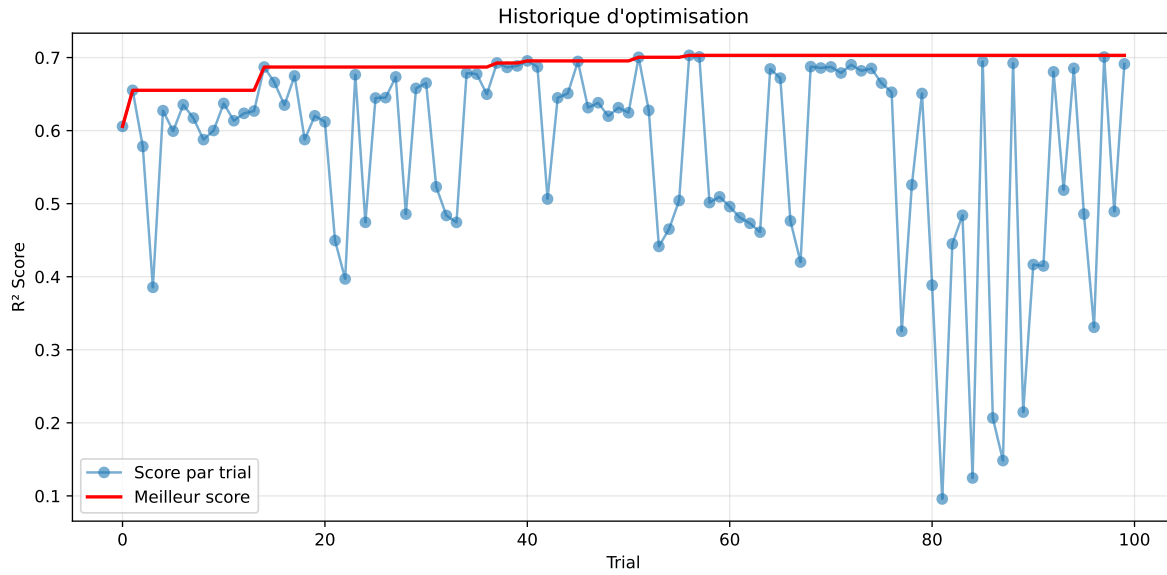
3.5 Model : LGBM

3.5.1 Gridsearch

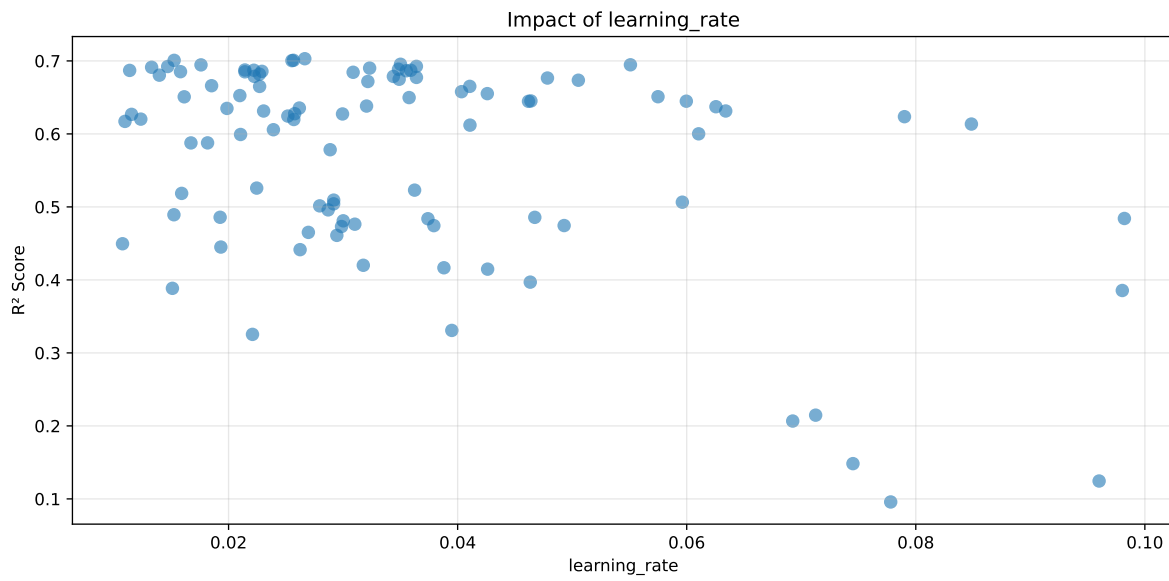
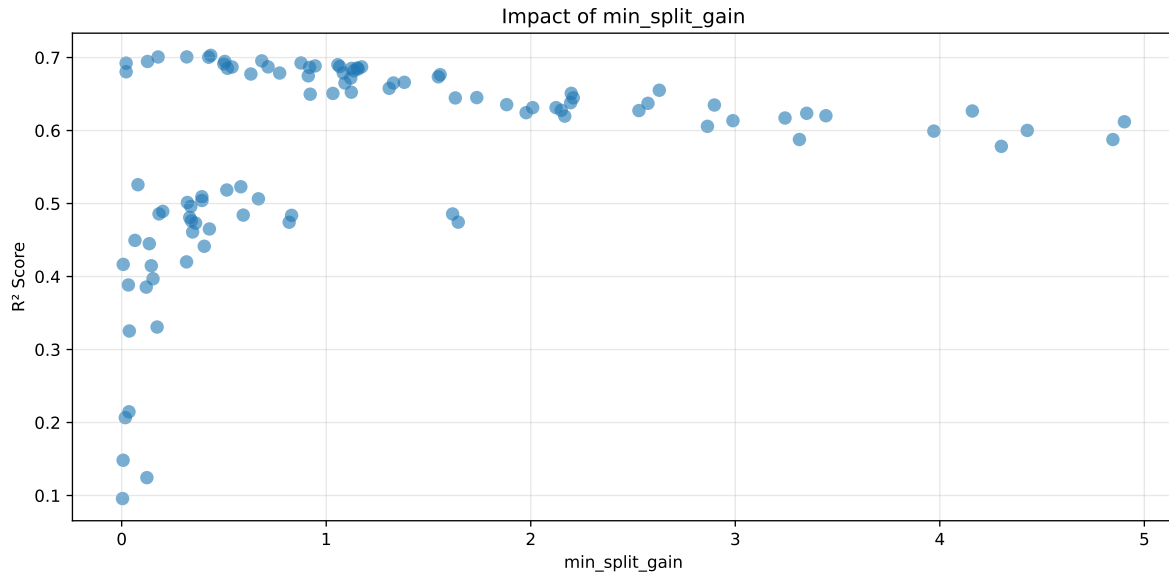
Distribution y_{train} vs y_{test} : y_{train} : mean=2.352, std=0.802, min=0.959, max=13.845
 y_{test} : mean=2.361, std=0.799, min=1.068, max=8.190

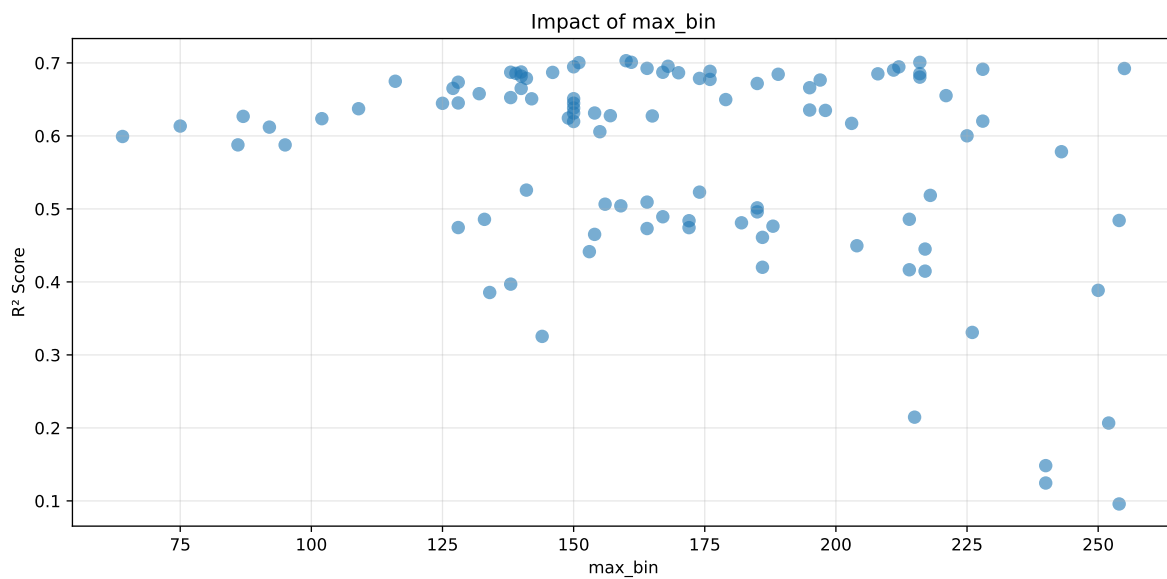
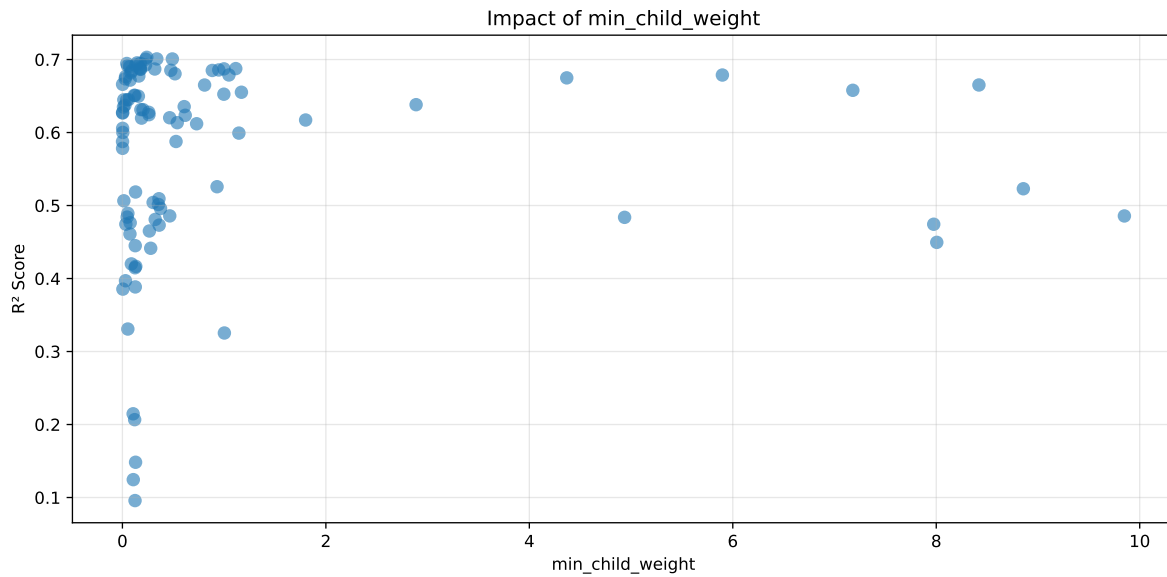
===== TOP Importance FEATURES =====

feature	importance
r_CO2	346
r_umidity	311
r_NH3	284
r_umidity_wl_8h_A_energy	273
r_NH3_wl_8h_A_energy	234
r_umidity_wl_8h_A_rms	189
r_temperature	174
r_temperature_wl_8h_A_energy_rel	168
r_NH3_wl_8h_D3_energy_rel	148
r_THI	144
r_temperature_wl_8h_A_rms	119
r_NH3_wl_8h_A_rms	116
r_umidity_wl_8h_D3_en- ergy_rel	115
r_NH3_wl_8h_D3_energy	107
r_NH3_wl_8h_D1_energy	107
r_tempera- ture_wl_8h_D3_energy	85
r_umidity_wl_8h_A_energy_rel	83
r_NH3_wl_8h_D2_energy	75
r_NH3_wl_8h_A_energy_rel	70
r_umidity_wl_8h_D2_energy	45



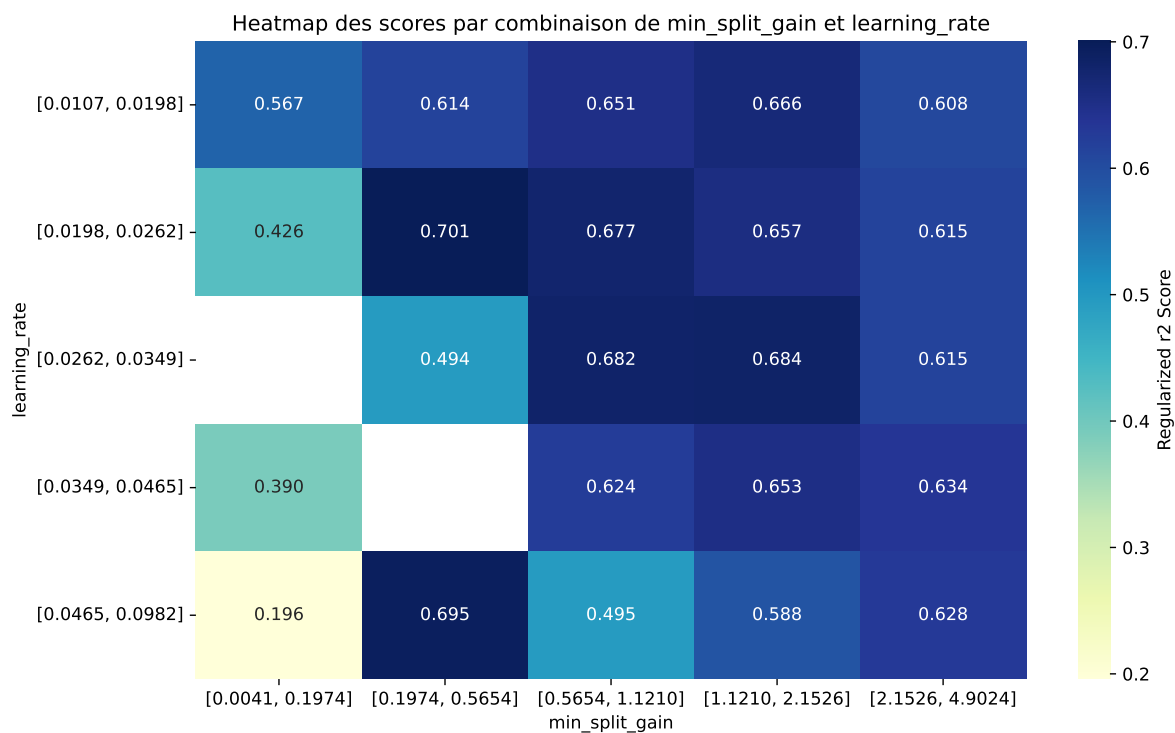
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'min_child_weight', 'max_bin']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7377	0.703	0.7018	0.009	0.703	0.7077	0.0047	1.0067	1.0493
2	0.7333	0.7008	0.7	0.0094	0.7008	0.7048	0.0039	1.0056	1.0464
3	0.7338	0.7008	0.7015	0.0097	0.7008	0.7183	0.0176	1.0251	1.0471
4	0.7334	0.7004	0.7039	0.0085	0.7004	0.7225	0.0221	1.0315	1.0471
5	0.7277	0.6954	0.6951	0.0072	0.6954	0.7064	0.011	1.0158	1.0465
6	0.7294	0.6947	0.699	0.0086	0.6947	0.721	0.0264	1.0379	1.05
7	0.7255	0.6946	0.6965	0.0098	0.6946	0.7086	0.014	1.0202	1.0446
8	0.7245	0.6925	0.694	0.0091	0.6925	0.718	0.0254	1.0367	1.0461
9	0.7228	0.6923	0.6925	0.0109	0.6923	0.7015	0.0092	1.0132	1.044
10	0.7241	0.6913	0.6969	0.0089	0.6913	0.7146	0.0233	1.0337	1.0474
---	---	---	---	---	---	---	---	---	---
100	0.8986	0.7648	0.7633	0.0076	0.0958	0.7861	0.0213	1.0279	1.1749

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.026655888343184476, 'reg__num_leaves': 45, 'reg__subsample': 0.5829419110442932, 'reg__colsample_bytree': 0.7698940665748341, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.1505955293741412, 'reg__reg_lambda': 31.206856199754434, 'reg__min_split_gain': 0.43549970061014226, 'reg__path_smooth': 2.866296993285685, 'reg__min_child_weight':

0.23943533609290957, 'reg__feature_fraction': 0.6218528659194031, 'reg__bagging_fraction': 0.6357033099002082, 'reg__bagging_freq': 5, 'reg__max_bin': 160, 'reg__min_data_per_group': 181}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.025667810604959613, 'reg__num_leaves': 43, 'reg__subsample': 0.7569195734663431, 'reg__colsample_bytree': 0.7739838428052055, 'reg__min_child_samples': 28, 'reg__reg_alpha': 1.2708637878681905, 'reg__reg_lambda': 45.09140902662772, 'reg__min_split_gain': 0.3181472427230192, 'reg__path_smooth': 3.180840718893178, 'reg__min_child_weight': 0.3381175851343129, 'reg__feature_fraction': 0.8062955359588908, 'reg__bagging_fraction': 0.6316663708253218, 'reg__bagging_freq': 5, 'reg__max_bin': 161, 'reg__min_data_per_group': 188}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.015249707182399205, 'reg__num_leaves': 51, 'reg__subsample': 0.704589181983097, 'reg__colsample_bytree': 0.5932167841088848, 'reg__min_child_samples': 31, 'reg__reg_alpha': 4.934035713210432, 'reg__reg_lambda': 40.35457666808086, 'reg__min_split_gain': 0.17819540102679032, 'reg__path_smooth': 2.277441665514711, 'reg__min_child_weight': 0.4907111485637966, 'reg__feature_fraction': 0.5760670346330231, 'reg__bagging_fraction': 0.746268553199744, 'reg__bagging_freq': 7, 'reg__max_bin': 216, 'reg__min_data_per_group': 163}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.025529851760053294, 'reg__num_leaves': 35, 'reg__subsample': 0.5878432311528371, 'reg__colsample_bytree': 0.839928744991095, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.344968955894311, 'reg__reg_lambda': 31.732789243131503, 'reg__min_split_gain': 0.42536980213928205, 'reg__path_smooth': 3.2919954314284117, 'reg__min_child_weight': 0.2285974929659044, 'reg__feature_fraction': 0.6291730023747001, 'reg__bagging_fraction': 0.6122869317528639, 'reg__bagging_freq': 6, 'reg__max_bin': 151, 'reg__min_data_per_group': 130}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.035012335055609835, 'reg__num_leaves': 24, 'reg__subsample': 0.7772651875995468, 'reg__colsample_bytree': 0.8988493239725521, 'reg__min_child_samples': 16, 'reg__reg_alpha': 1.3876307804326782, 'reg__reg_lambda': 34.57374728096022, 'reg__min_split_gain': 0.685068451024814, 'reg__path_smooth': 3.413557137075424, 'reg__min_child_weight': 0.14389406339375452, 'reg__feature_fraction': 0.6662646457761731, 'reg__bagging_fraction': 0.6740141441671637, 'reg__bagging_freq': 6, 'reg__max_bin': 168, 'reg__min_data_per_group': 138}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.0550860314926493, 'reg__num_leaves': 32, 'reg__subsample': 0.5379136045955587, 'reg__colsample_bytree': 0.6741259898685996, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.2275182585780415, 'reg__reg_lambda': 12.040921285308068, 'reg__min_split_gain': 0.504316805427359, 'reg__path_smooth': 3.2316612766894255, 'reg__min_child_weight':

0.17589363667939442, 'reg__feature_fraction': 0.7746534284116946, 'reg__bagging_fraction': 0.5955696470567241, 'reg__bagging_freq': 6, 'reg__max_bin': 150, 'reg__min_data_per_group': 126}

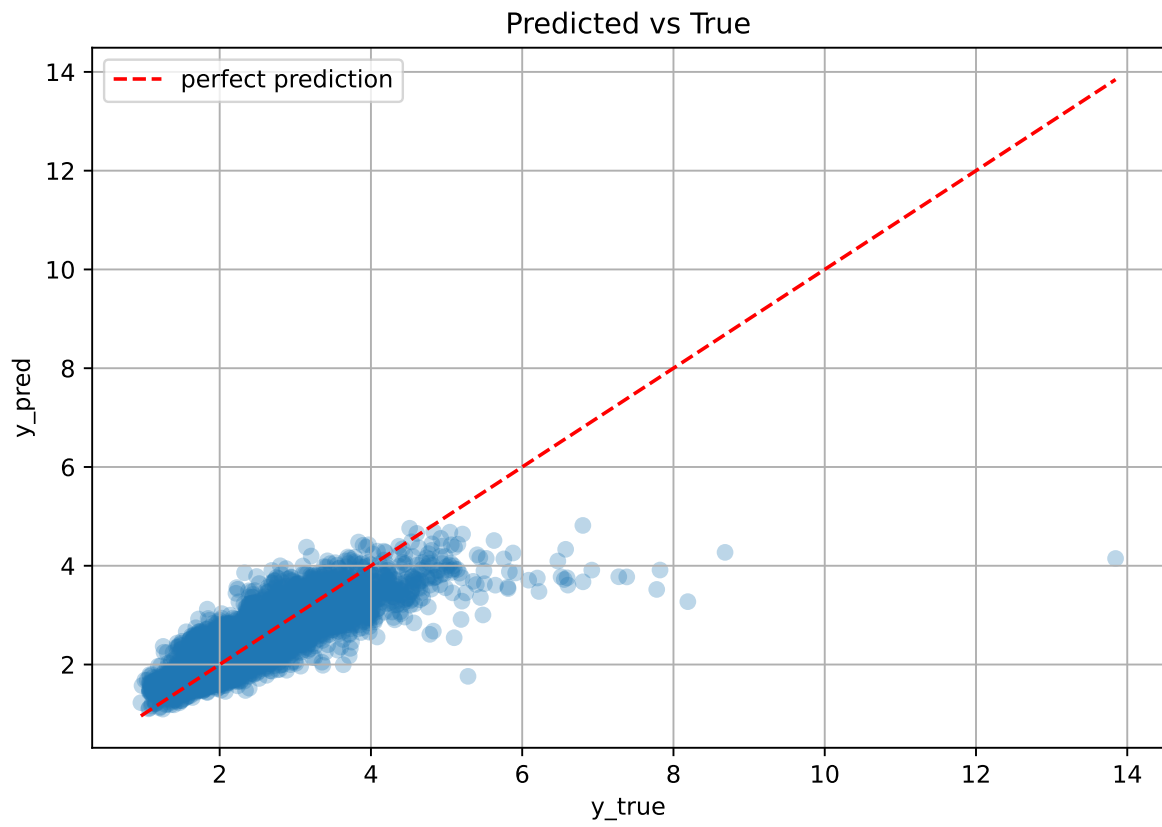
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.017592001508719576, 'reg__num_leaves': 47, 'reg__subsample': 0.6954384734703802, 'reg__colsample_bytree': 0.862917064498122, 'reg__min_child_samples': 21, 'reg__reg_alpha': 5.665973473138739, 'reg__reg_lambda': 37.22517254572854, 'reg__min_split_gain': 0.126371660520745, 'reg__path_smooth': 3.4235951055734373, 'reg__min_child_weight': 0.043010998416996044, 'reg__feature_fraction': 0.8642023487991822, 'reg__bagging_fraction': 0.7227667704482874, 'reg__bagging_freq': 7, 'reg__max_bin': 212, 'reg__min_data_per_group': 156}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.036399515183309986, 'reg__num_leaves': 34, 'reg__subsample': 0.5460341250947796, 'reg__colsample_bytree': 0.7672586178325096, 'reg__min_child_samples': 15, 'reg__reg_alpha': 1.8627771466286465, 'reg__reg_lambda': 33.50141138783532, 'reg__min_split_gain': 0.876856629889267, 'reg__path_smooth': 3.5740901563610614, 'reg__min_child_weight': 0.22732419318206135, 'reg__feature_fraction': 0.6694642554931873, 'reg__bagging_fraction': 0.6617319046538782, 'reg__bagging_freq': 6, 'reg__max_bin': 164, 'reg__min_data_per_group': 136}

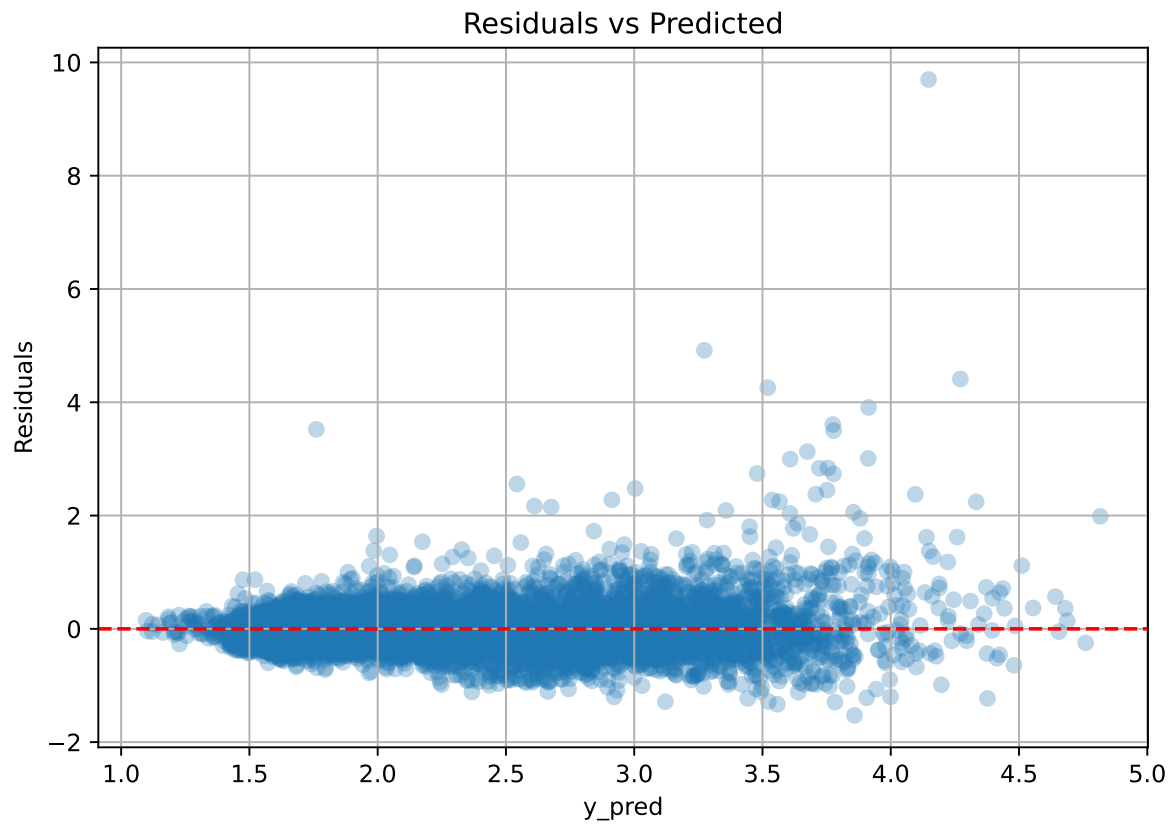
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.014685007442934403, 'reg__num_leaves': 47, 'reg__subsample': 0.5331458873893591, 'reg__colsample_bytree': 0.6151570307757421, 'reg__min_child_samples': 22, 'reg__reg_alpha': 0.4922743476562508, 'reg__reg_lambda': 37.76544145432298, 'reg__min_split_gain': 0.0217315764643089, 'reg__path_smooth': 2.2030351369968217, 'reg__min_child_weight': 0.12881449682716137, 'reg__feature_fraction': 0.8678528911198523, 'reg__bagging_fraction': 0.6665034807176299, 'reg__bagging_freq': 7, 'reg__max_bin': 255, 'reg__min_data_per_group': 166}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.013281510649975967, 'reg__num_leaves': 50, 'reg__subsample': 0.7103668047171, 'reg__colsample_bytree': 0.8238728207323445, 'reg__min_child_samples': 23, 'reg__reg_alpha': 6.885715345488229, 'reg__reg_lambda': 40.89324671125208, 'reg__min_split_gain': 0.49932741038546574, 'reg__path_smooth': 2.193403443115766, 'reg__min_child_weight': 0.050966748072058136, 'reg__feature_fraction': 0.8892719852499127, 'reg__bagging_fraction': 0.7535918197113386, 'reg__bagging_freq': 6, 'reg__max_bin': 228, 'reg__min_data_per_group': 108}

3.5.2 Scatter



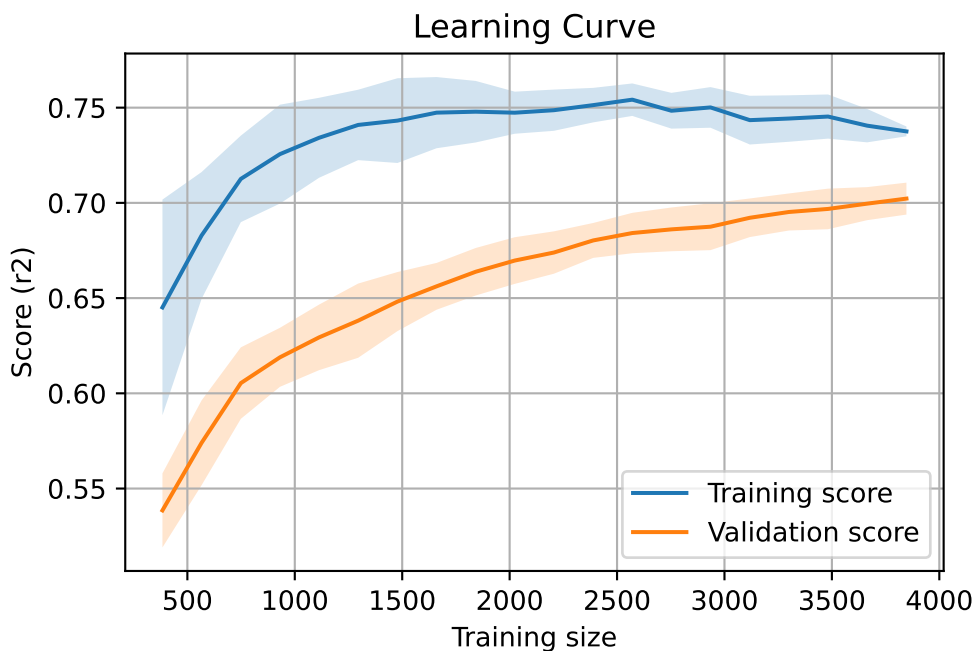
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7852, `Len(training)` : 5497,

`Len(training_real)` : 3848, `Len(test_of_training)` : 1649, `Len(test)` : 2355



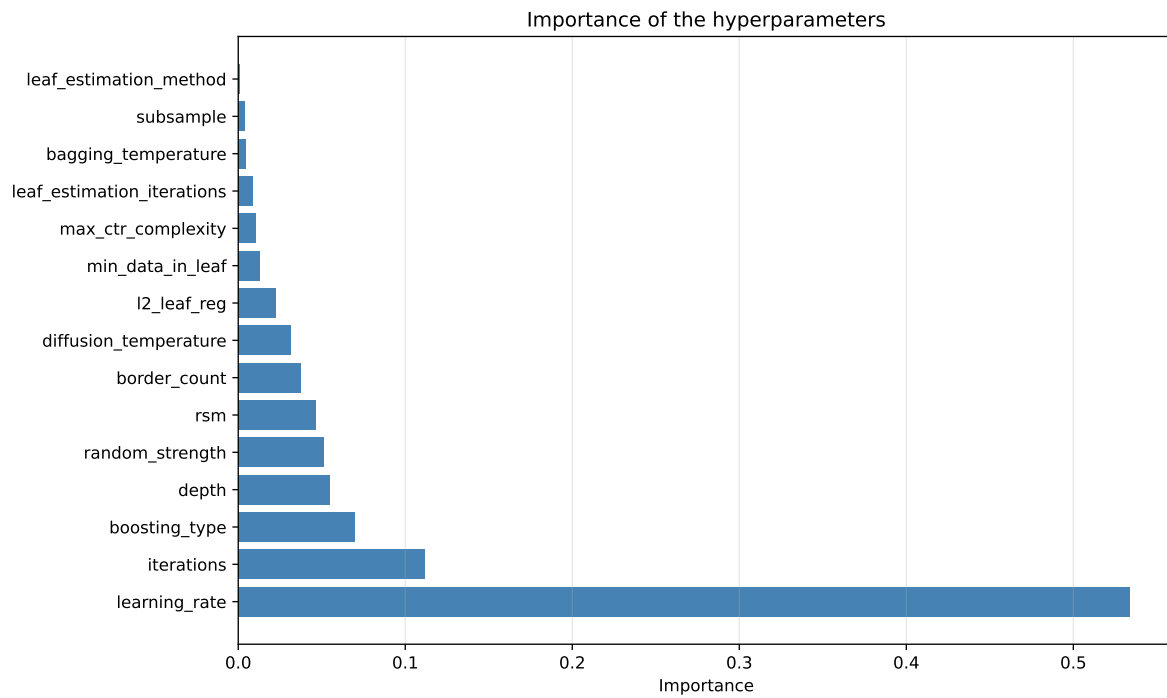
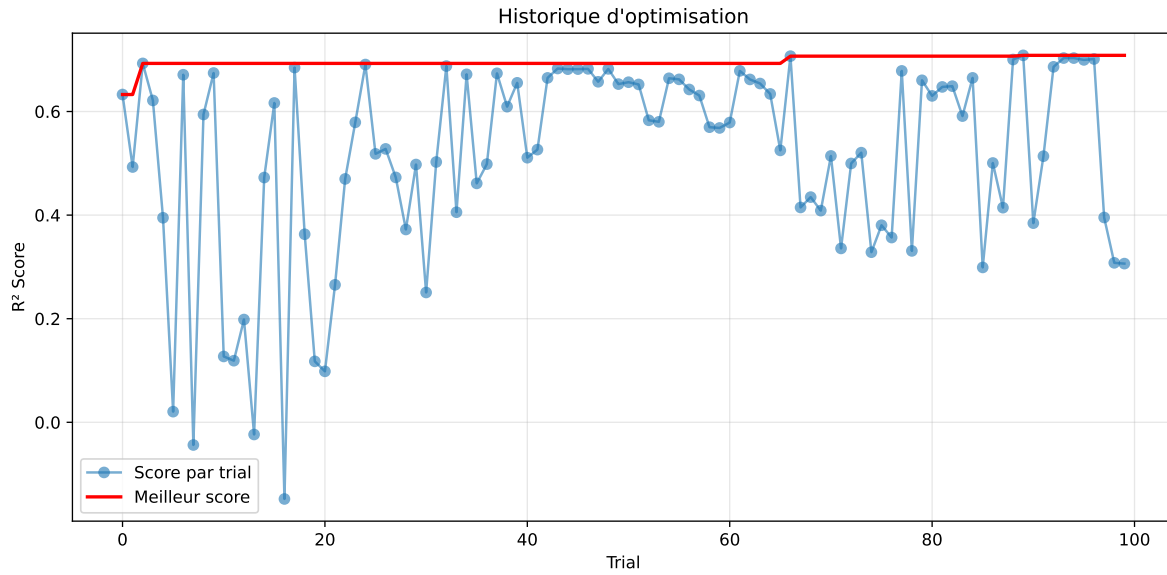
3.6 Model : CatBoost

3.6.1 Gridsearch

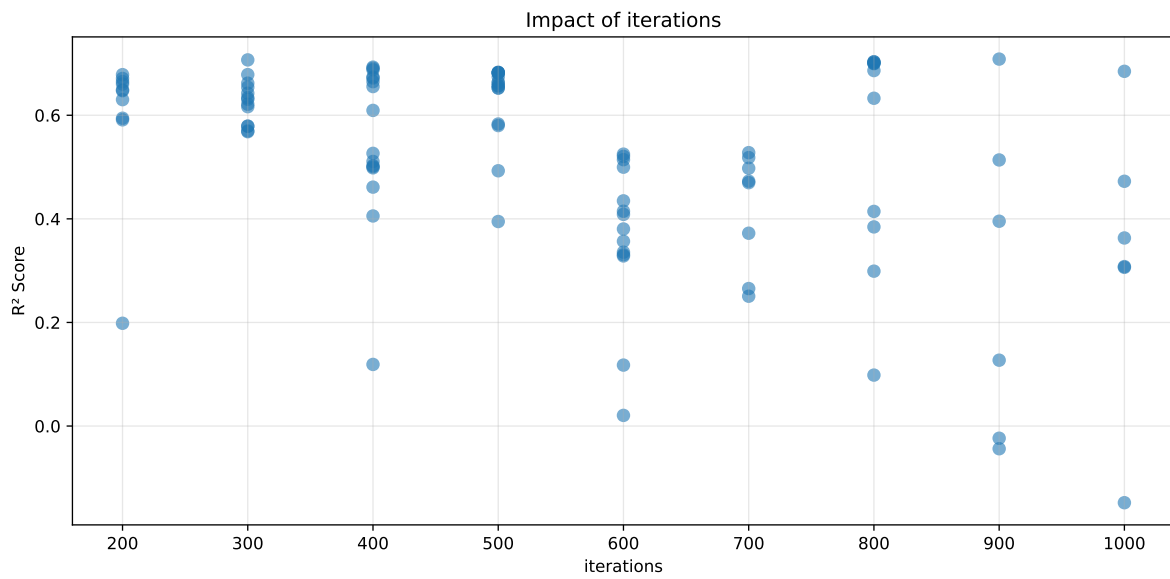
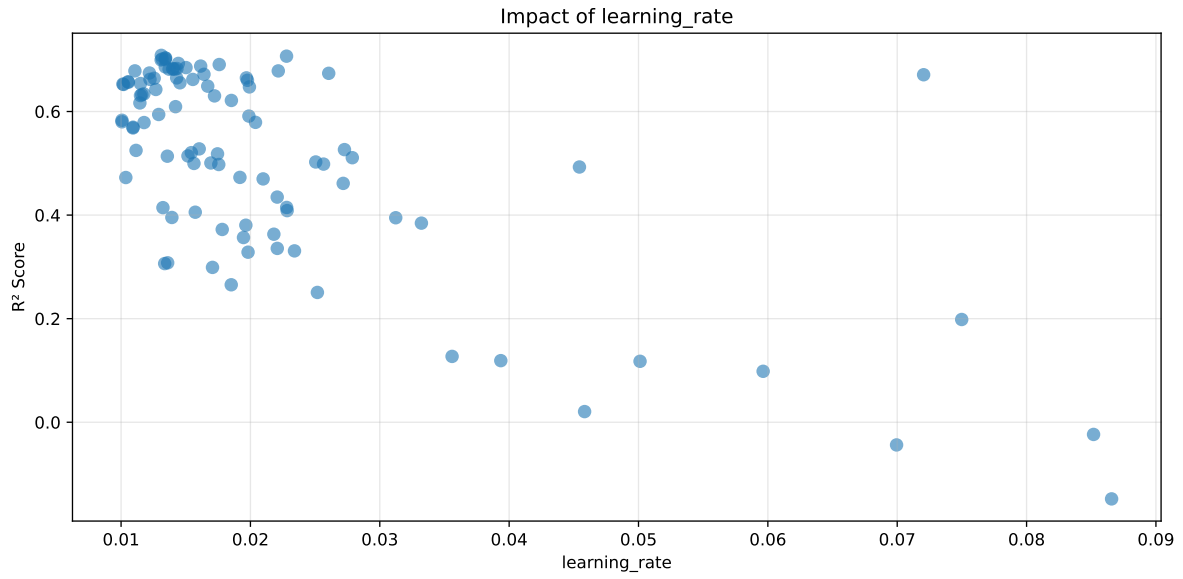
Distribution y_train vs y_test: y_train: mean=2.352, std=0.802, min=0.959, max=13.845
y_test: mean=2.361, std=0.799, min=1.068, max=8.190

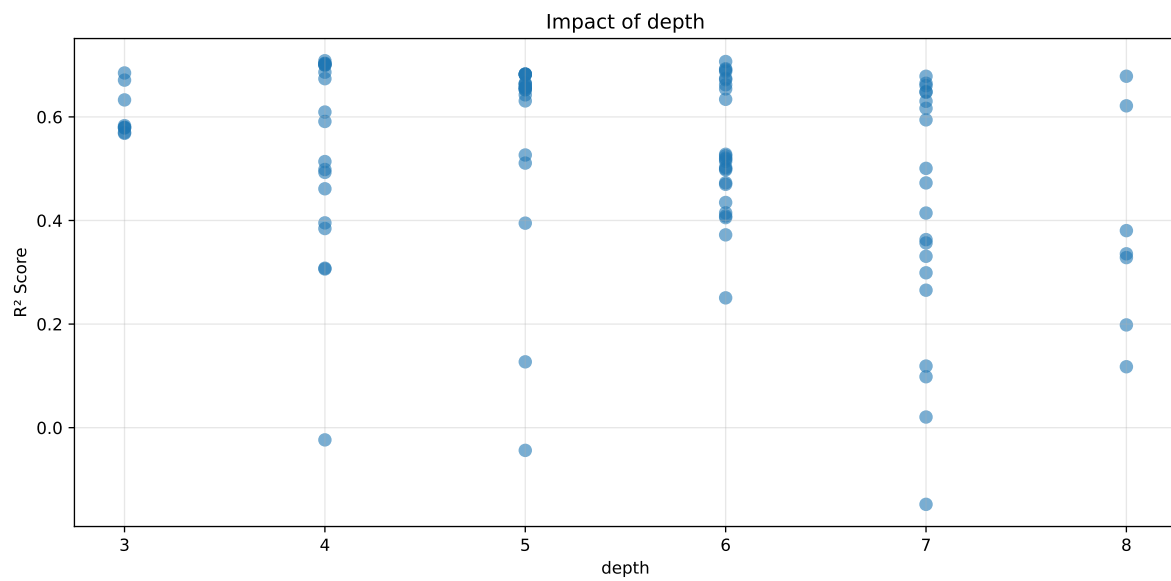
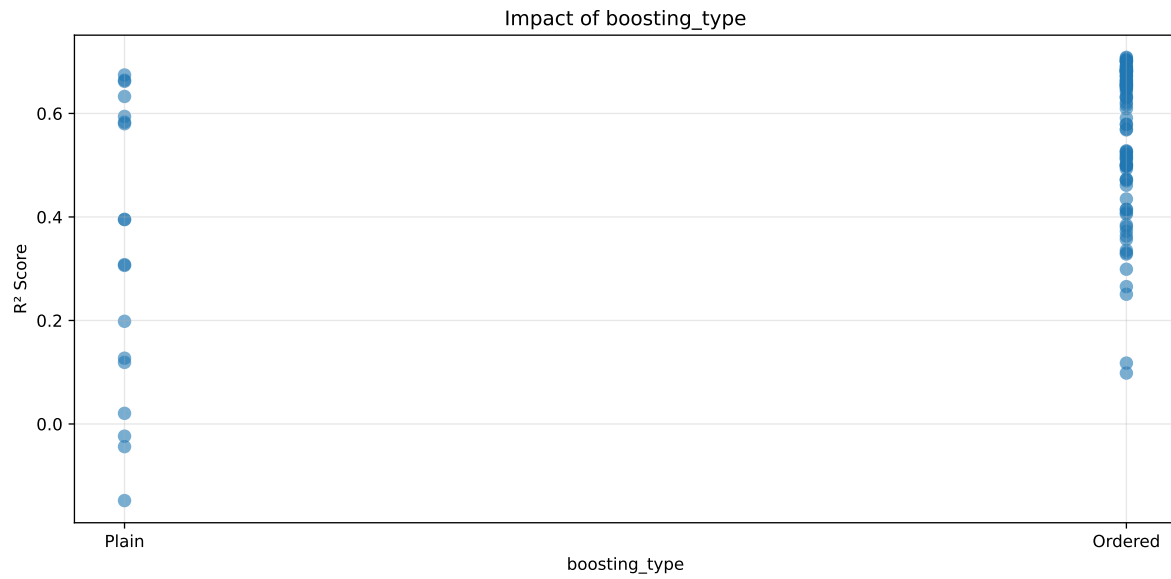
===== TOP Importance FEATURES =====

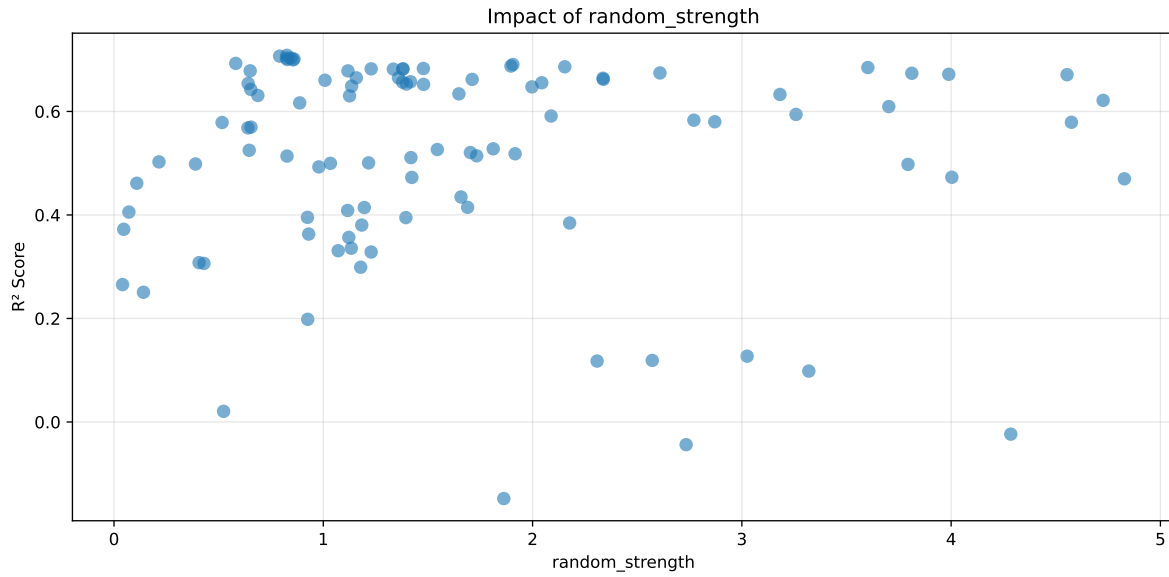
feature importance	r_umidity	22.473625	r_temperature_wl_8h_A_energy_rel	15.625230	
r_umidity_wl_8h_A_energy	9.419532	r_CO2	8.165684	r_NH3	7.724994
r_temperature	5.632843	r_NH3_wl_8h_A_energy	4.614958	r_THI	3.434113
r_NH3_wl_8h_D1_energy	3.414082	r_NH3_wl_8h_D3_energy	3.066192	r_umidity_wl_8h_A_rms	2.496574
r_NH3_wl_8h_D3_energy_rel	2.199895	r_NH3_wl_8h_D2_energy	1.970010	r_NH3_wl_8h_A_rms	1.866534
r_temperature_wl_8h_D3_energy	1.760707	r_temperature_wl_8h_A_rms	1.622282	r_umidity_wl_8h_A_energy_rel	1.403024
r_umidity_wl_8h_D3_energy_rel	1.220908	r_NH3_wl_8h_A_energy_rel	1.053958	r_umidity_wl_8h_D2_energy	0.834854



Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'boosting_type', 'depth', 'random_strength']

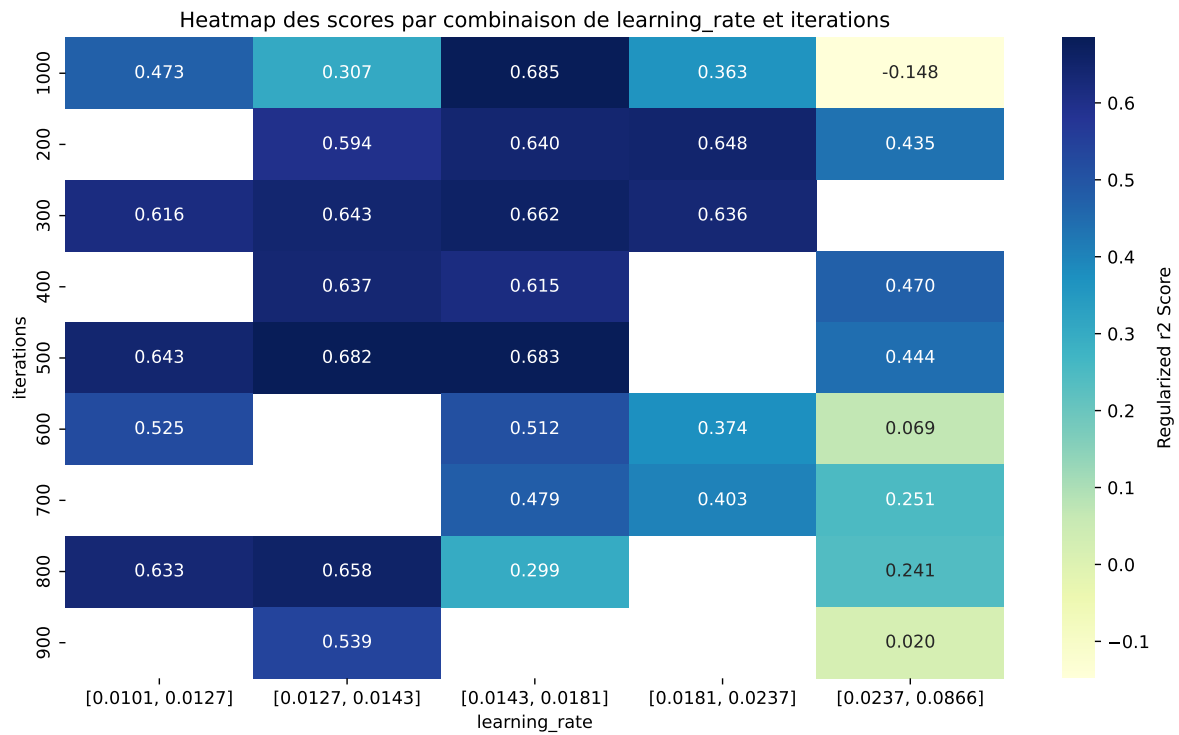






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7431	0.7084	0.7078	0.0093	0.7084	0.7152	0.0068	1.0096	1.049
2	0.7417	0.7069	0.7072	0.0101	0.7069	0.7174	0.0106	1.0149	1.0493
3	0.7365	0.7032	0.7024	0.0095	0.7032	0.706	0.0028	1.004	1.0473
4	0.7368	0.7031	0.7019	0.0094	0.7031	0.7053	0.0021	1.003	1.0479
5	0.7336	0.7014	0.6994	0.0103	0.7014	0.705	0.0036	1.0051	1.0459
6	0.7296	0.7004	0.6996	0.0106	0.7004	0.7052	0.0049	1.007	1.0417
7	0.7299	0.6996	0.699	0.0105	0.6996	0.7054	0.0057	1.0082	1.0433
8	0.7179	0.6929	0.6933	0.0111	0.6929	0.6997	0.0069	1.0099	1.0361
9	0.7179	0.6907	0.6885	0.0115	0.6907	0.6978	0.0071	1.0103	1.0394
10	0.7145	0.6877	0.6847	0.0114	0.6877	0.6942	0.0065	1.0095	1.0389
---	---	---	---	---	---	---	---	---	---
100	0.9927	0.8026	0.7961	0.0102	-0.148	0.8165	0.0139	1.0174	1.2369

Hyperparameters:

Rank 1: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.013111209935731357, 'reg__l2_leaf_reg': 6.995633826456855, 'reg__bagging_temperature': 2.8894520267251638, 'reg__random_strength': 0.8258736613914488, 'reg__subsample': 0.8434193305133639, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6958469018670237, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3118.4534251047066}

Rank 2: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.022791836158925997, 'reg__l2_leaf_reg': 10.71125059647187, 'reg__bagging_temperature': 3.1607328415911615, 'reg__random_strength': 0.7912602766697376, 'reg__subsample': 0.7211780181281475, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8627581202060955, 'reg__border_count': 109, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4321.555599099339}

Rank 3: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013440303848925809, 'reg__l2_leaf_reg': 3.718614939626086, 'reg__bagging_temperature': 0.20464296439401372, 'reg__random_strength': 0.8241892899458071, 'reg__subsample': 0.7437547802865279, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8391218020455199, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2454.3629001676563}

Rank 4: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013427179462120792, 'reg__l2_leaf_reg': 3.1471652586847645, 'reg__bagging_temperature': 0.19608776086596855, 'reg__random_strength': 0.8447747456045309, 'reg__subsample': 0.743152377935531, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 7, 'reg__rsm':

0.8863360350233174, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2441.1930152941886}

Rank 5: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013179769776186467, 'reg__l2_leaf_reg': 2.9502642365359897, 'reg__bagging_temperature': 4.694930554816519, 'reg__random_strength': 0.8599803596783312, 'reg__subsample': 0.8214542557926331, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8877822232922233, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1902.3375447761273}

Rank 6: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013407774552136912, 'reg__l2_leaf_reg': 7.785916986626682, 'reg__bagging_temperature': 3.897440899577399, 'reg__random_strength': 0.8279652757859628, 'reg__subsample': 0.8316715218149043, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.703205583448393, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2037.1585304531284}

Rank 7: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013097111830530254, 'reg__l2_leaf_reg': 2.9475520302004607, 'reg__bagging_temperature': 4.724283651611496, 'reg__random_strength': 0.8555068674924042, 'reg__subsample': 0.834723305586848, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6537655577496206, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2509.622339717839}

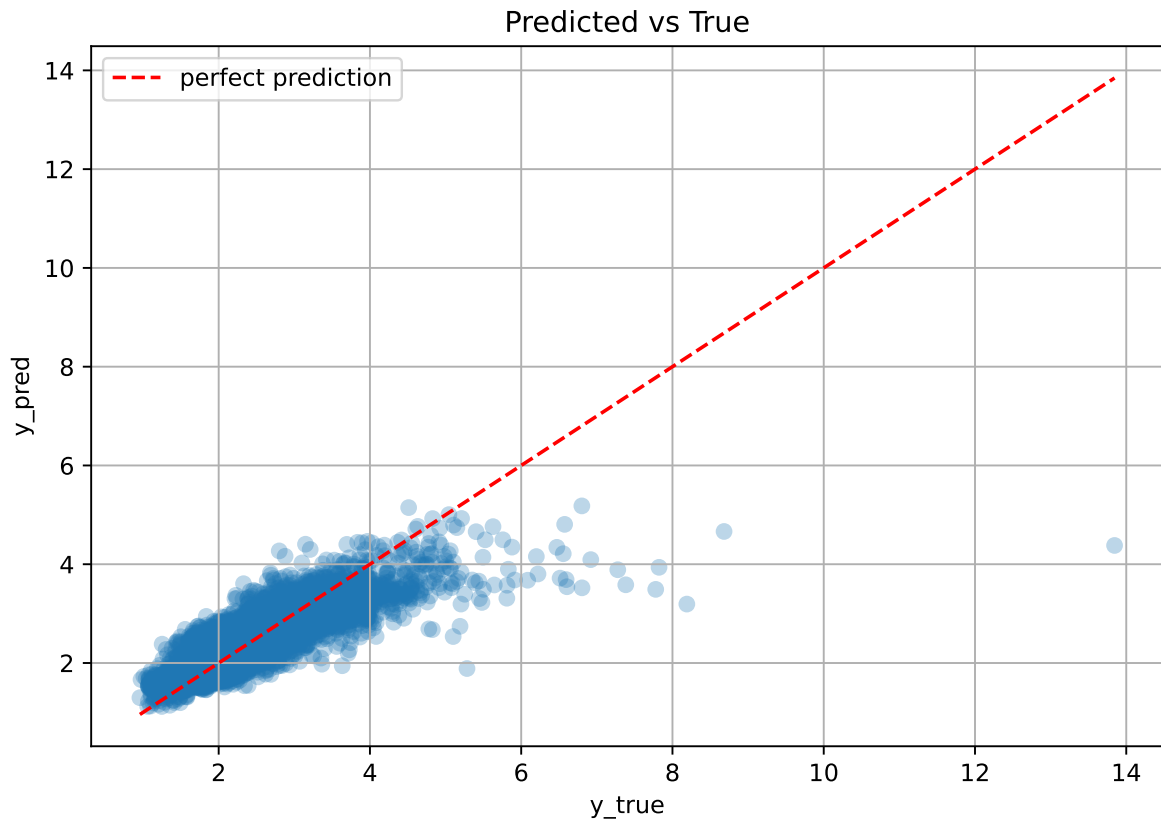
Rank 8: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.014428330393882222, 'reg__l2_leaf_reg': 21.700661354484335, 'reg__bagging_temperature': 3.375490521441527, 'reg__random_strength': 0.5817497505903252, 'reg__subsample': 0.706823300134473, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6057130159274473, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1433.8528793466144}

Rank 9: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.017587925895252533, 'reg__l2_leaf_reg': 1.4183521461843576, 'reg__bagging_temperature': 4.721730814677478, 'reg__random_strength': 1.905972842754843, 'reg__subsample': 0.5041180771389066, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8875043888655791, 'reg__border_count': 99, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2107.9930926884003}

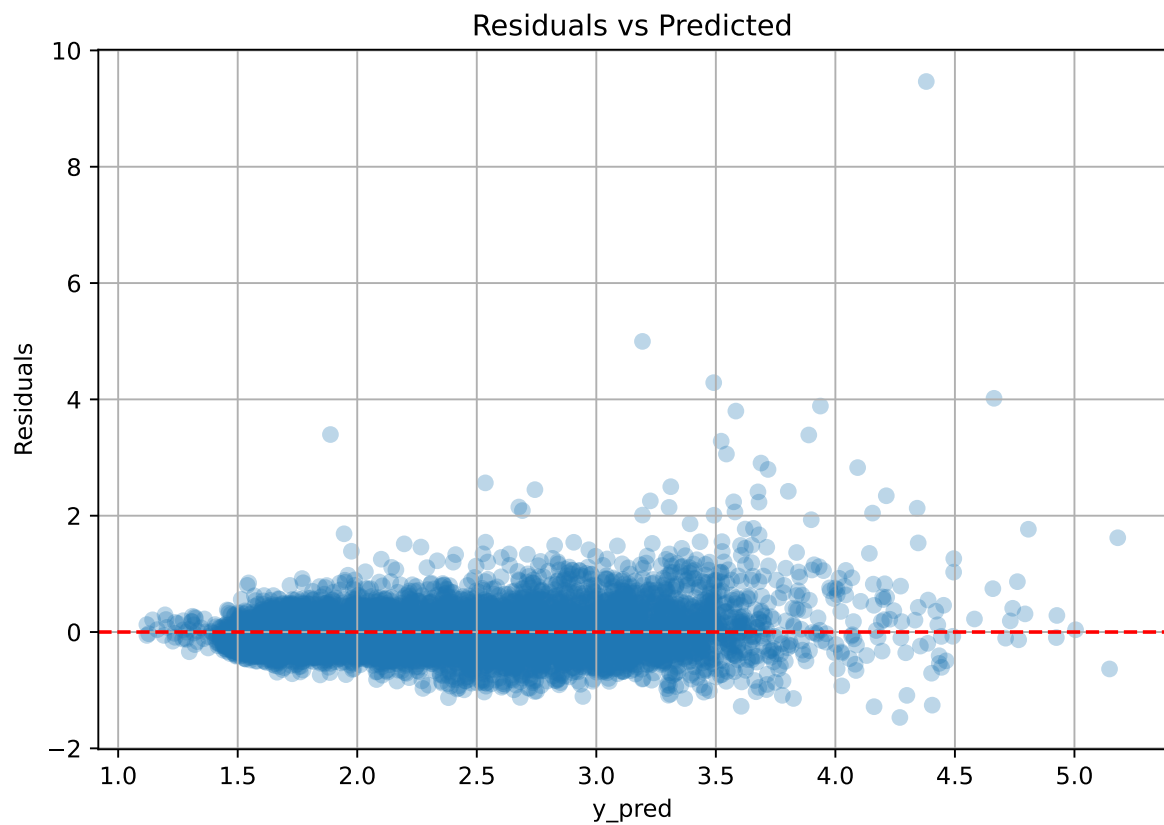
Rank 10: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.016152789100583795, 'reg__l2_leaf_reg': 3.218907471016175, 'reg__bagging_temperature': 1.3096747655323924,

'reg__random_strength': 1.8965677628292374, 'reg__subsample': 0.5953202567826674,
'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':
0.8988212362318132, 'reg__border_count': 112, 'reg__leaf_estimation_method': 'Newton',
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_tempera-
ture': 1609.8434710236966}

3.6.2 Scatter



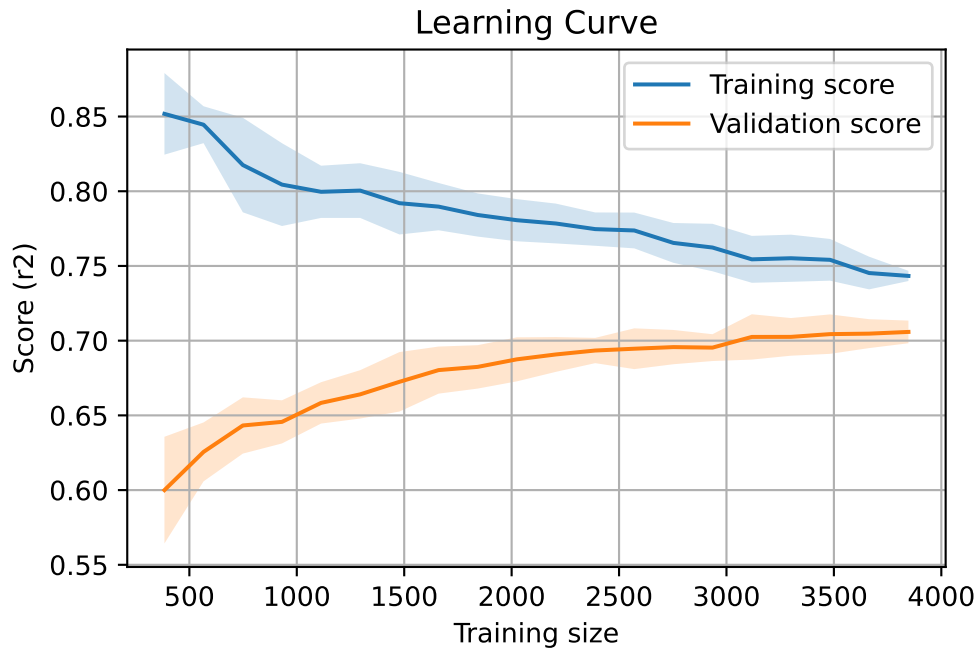
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7852, `Len(training)` : 5497,

`Len(training_real)` : 3848, `Len(test_of_training)` : 1649, `Len(test)` : 2355



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.532	0.547	0.007	0.528	0.53	-0.017	0.969	0.972	nan
LGBM	0.738	0.703	0.009	0.702	0.708	0.005	1.007	1.049	0.703
CatBoost	0.743	0.708	0.009	0.708	0.715	0.007	1.01	1.049	0.7084

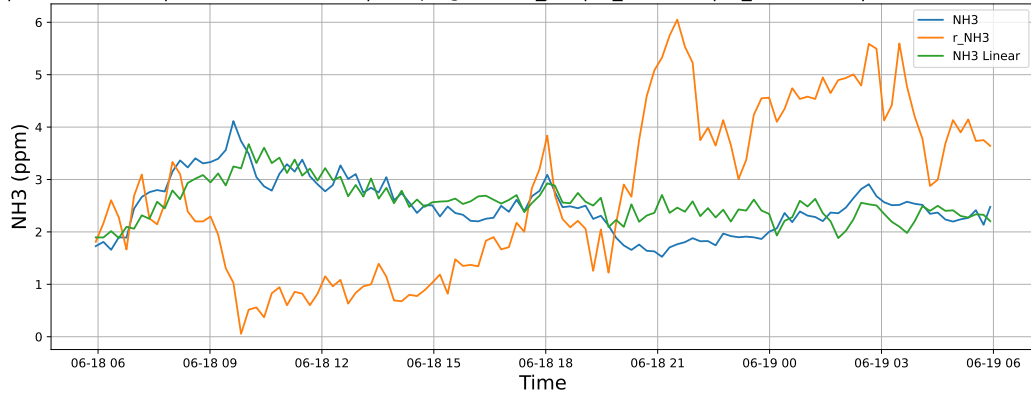
4 Plot

4.1 Standalone plots

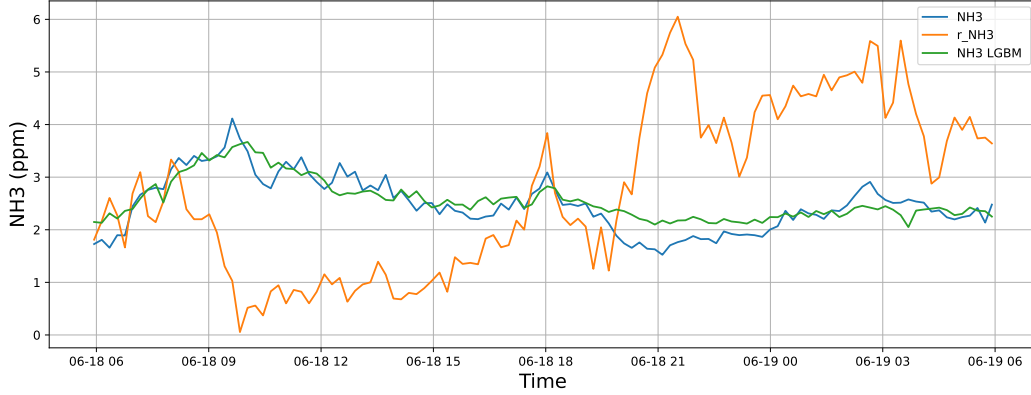
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

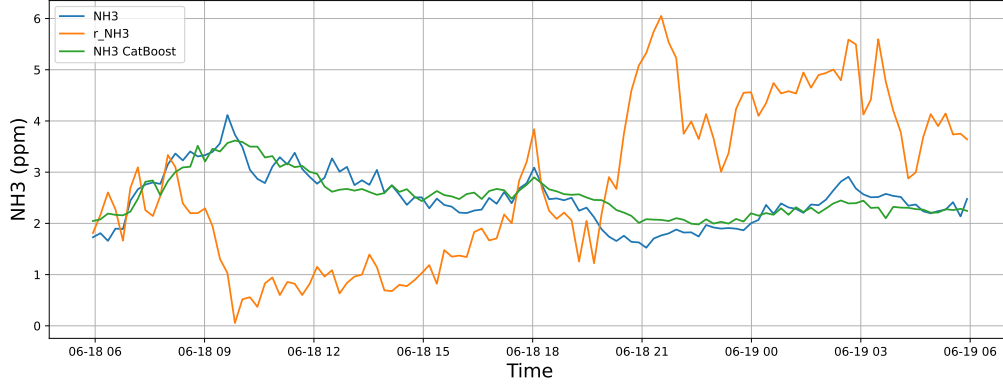
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

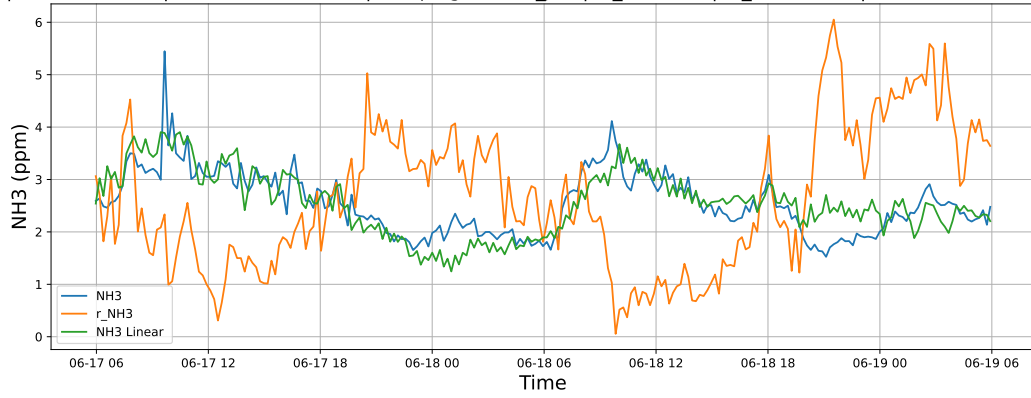


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

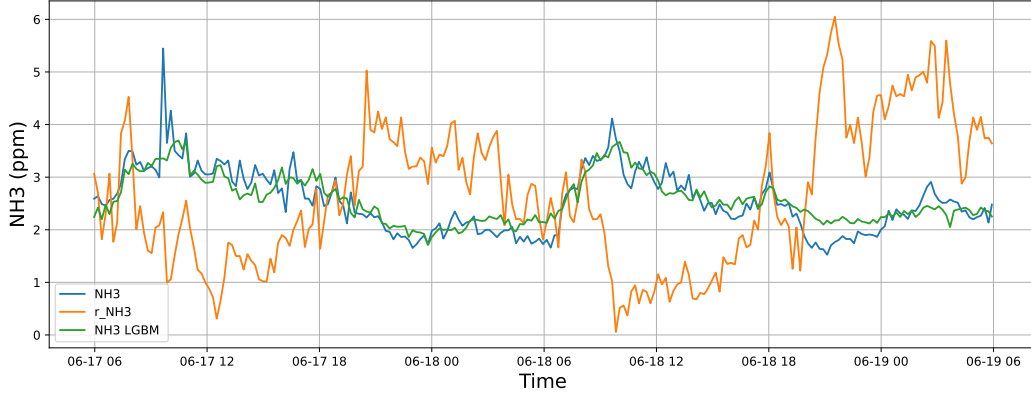


4.1.1.2 Time window : 48

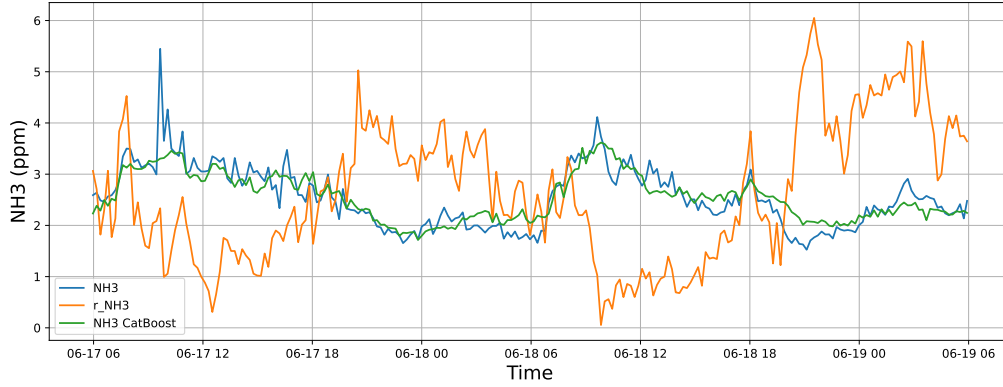
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

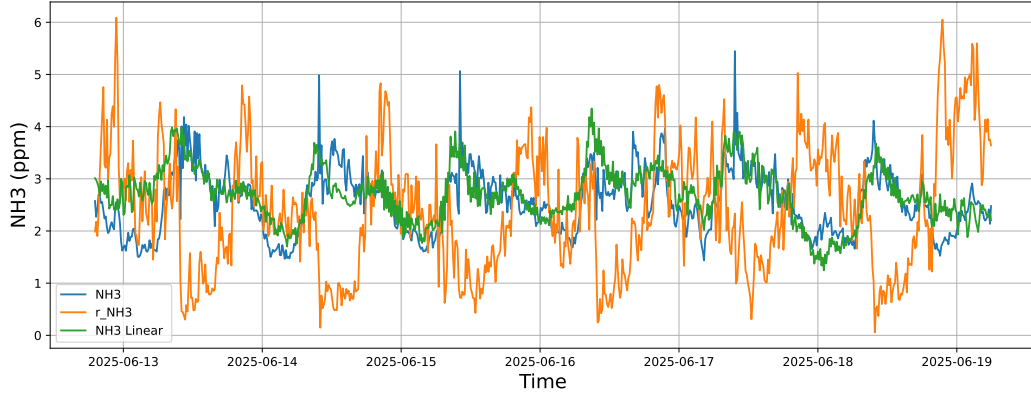


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

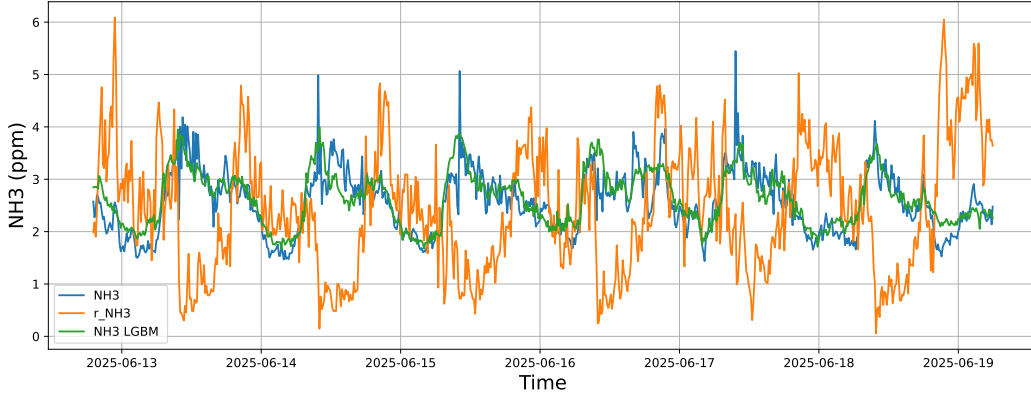


4.1.1.3 Time window : ALL

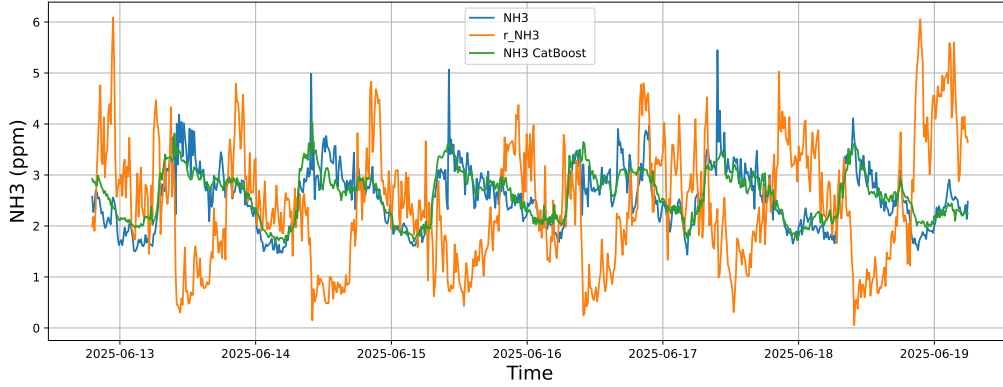
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



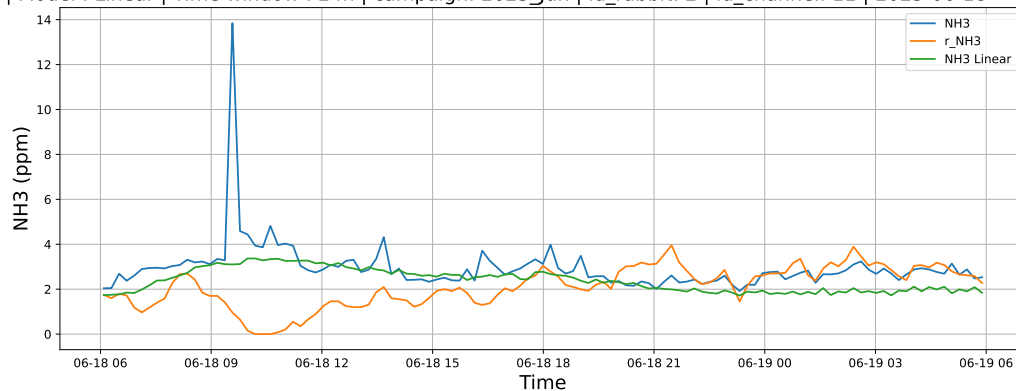
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



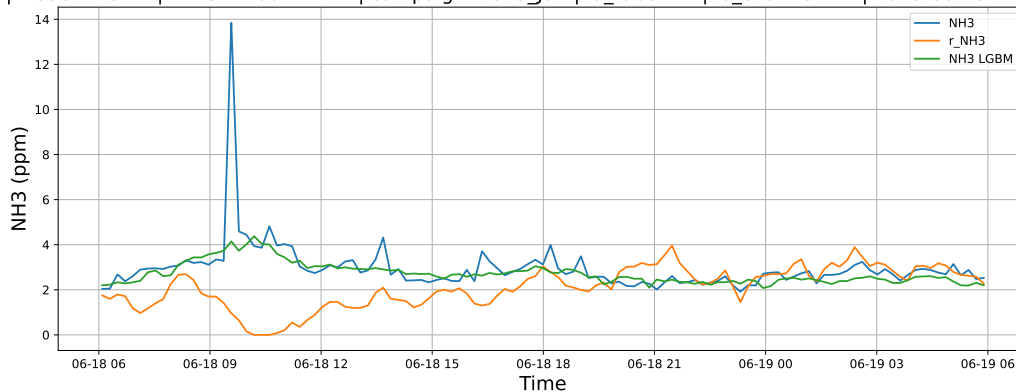
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

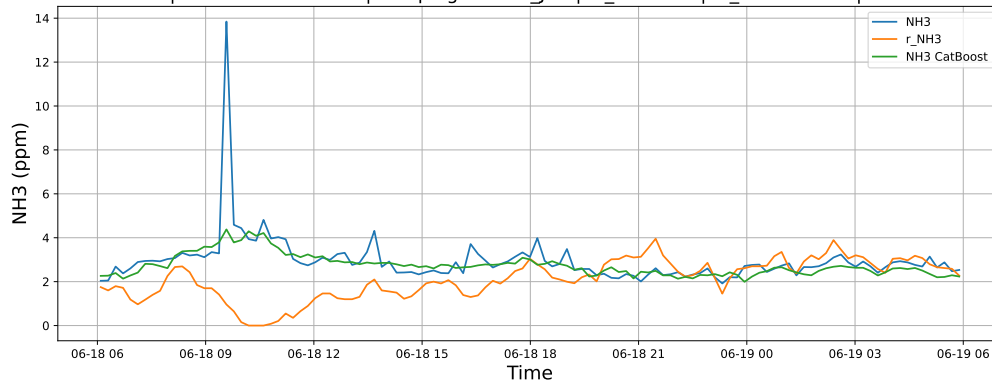
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

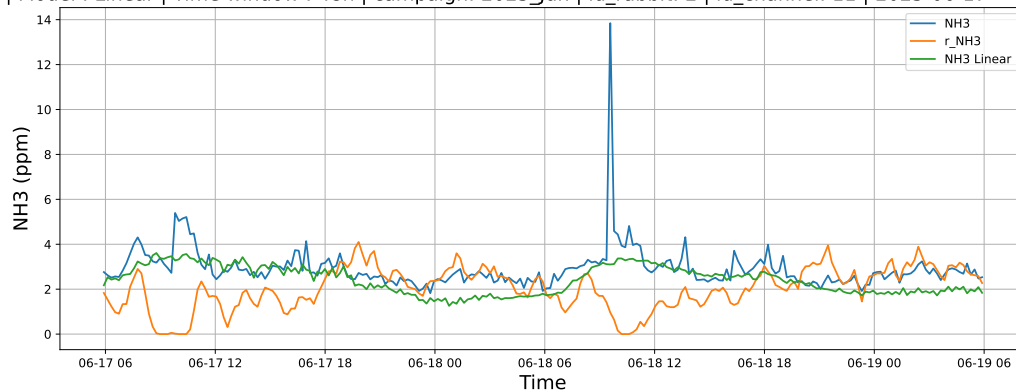


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

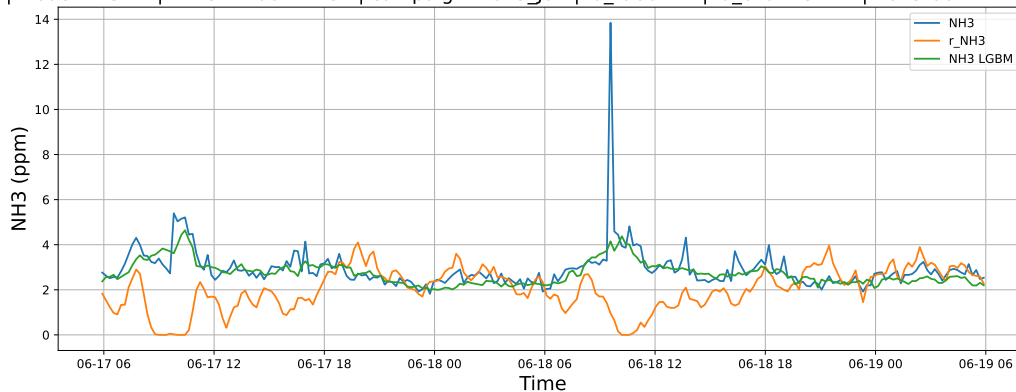


4.1.2.2 Time window : 48

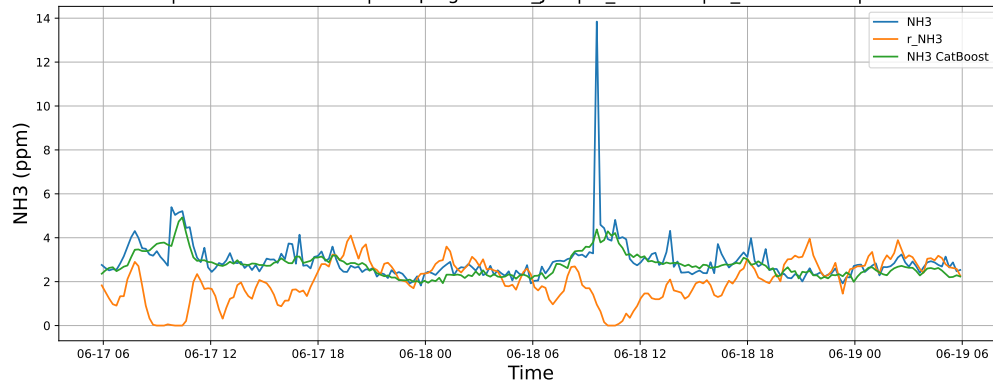
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

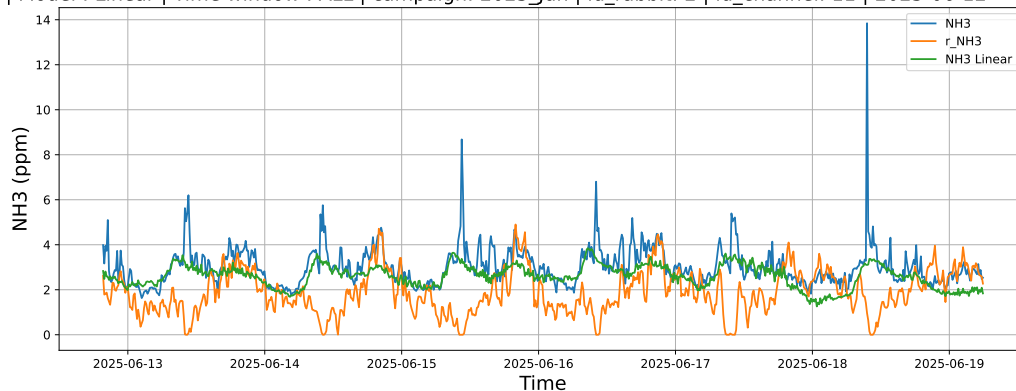


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

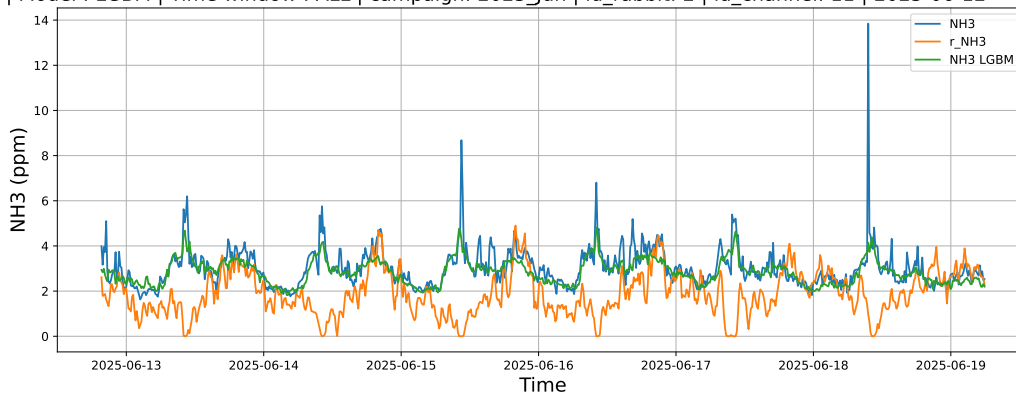


4.1.2.3 Time window : ALL

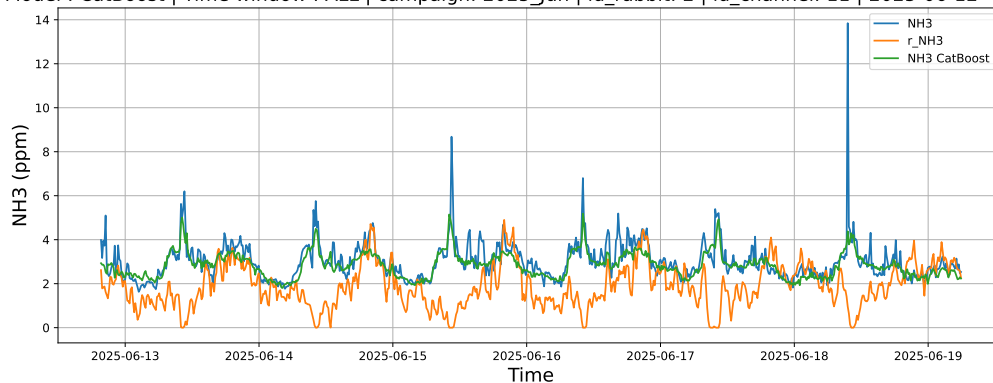
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



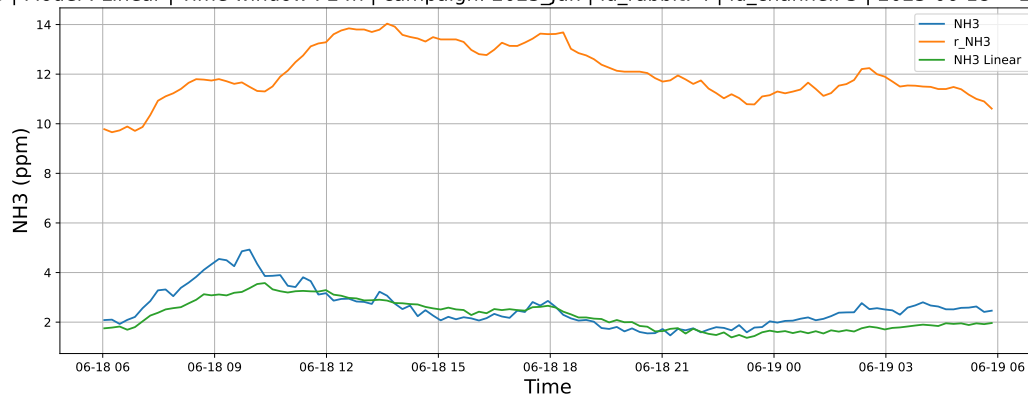
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



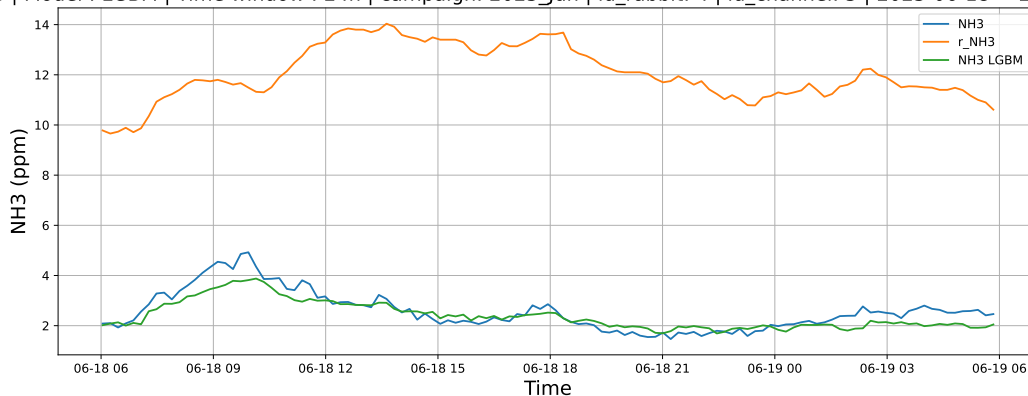
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

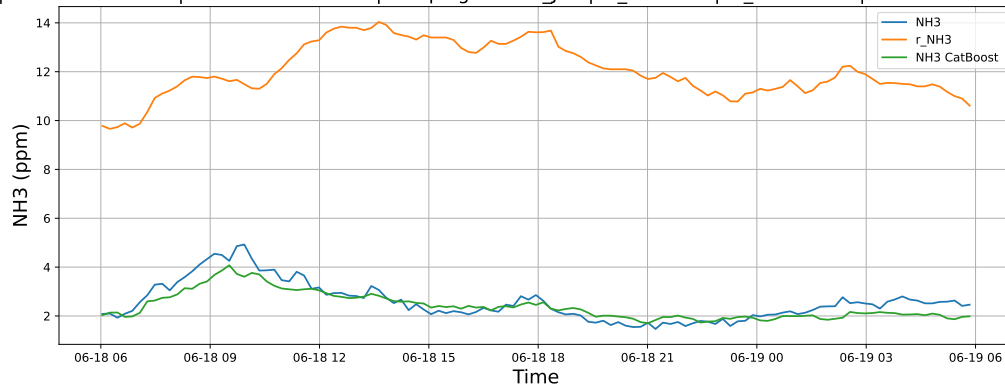
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

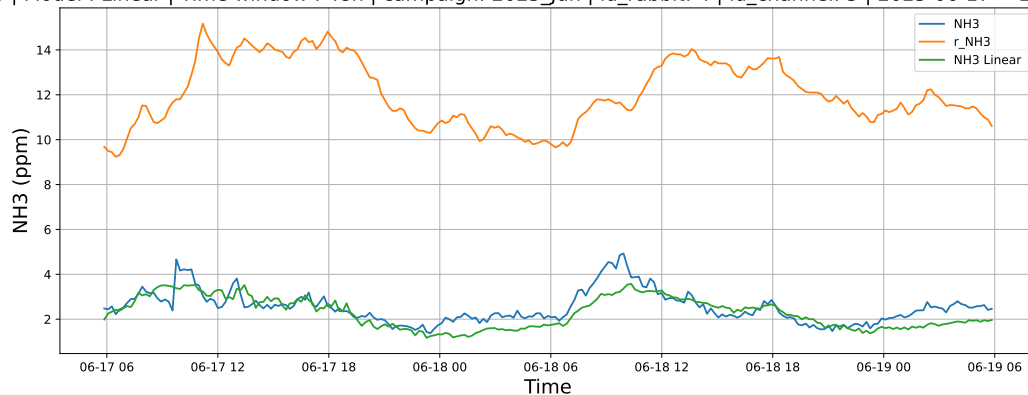


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

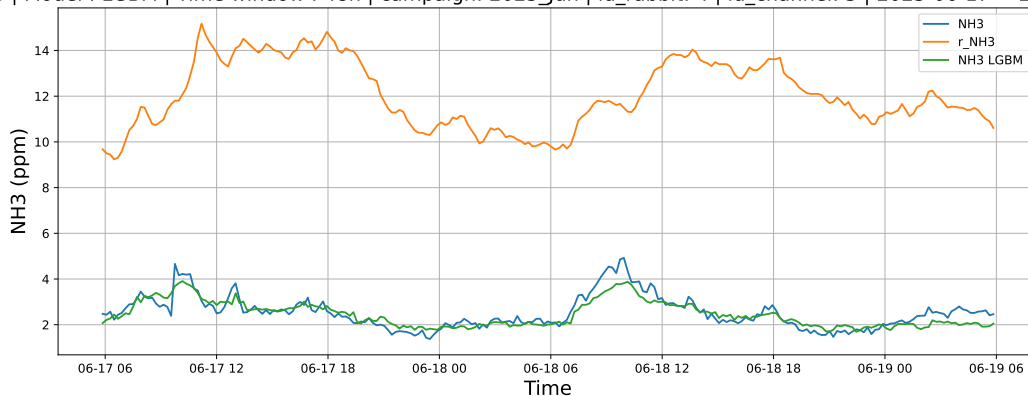


4.1.3.2 Time window : 48

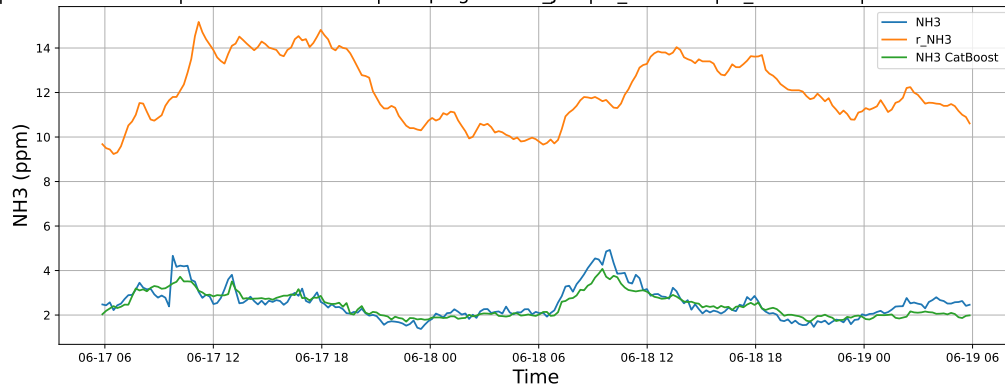
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

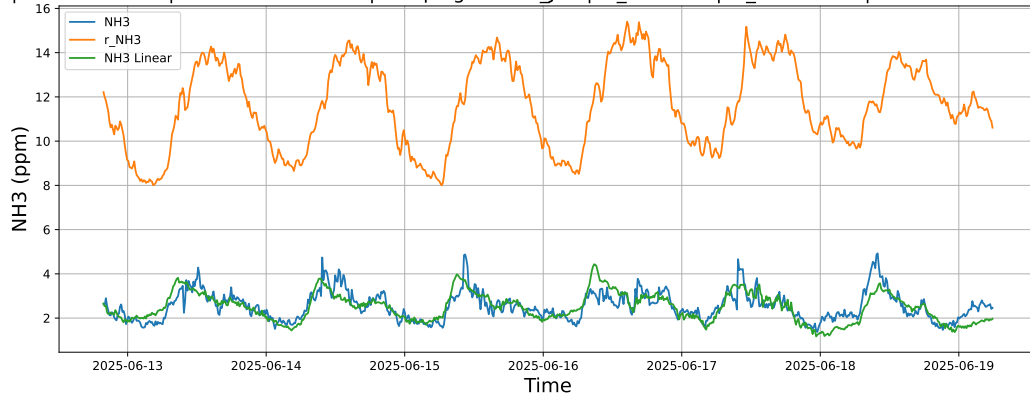


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

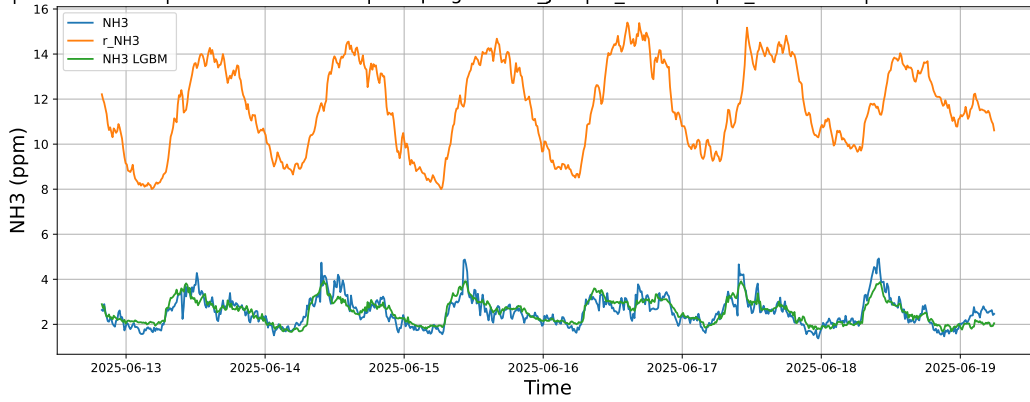


4.1.3.3 Time window : ALL

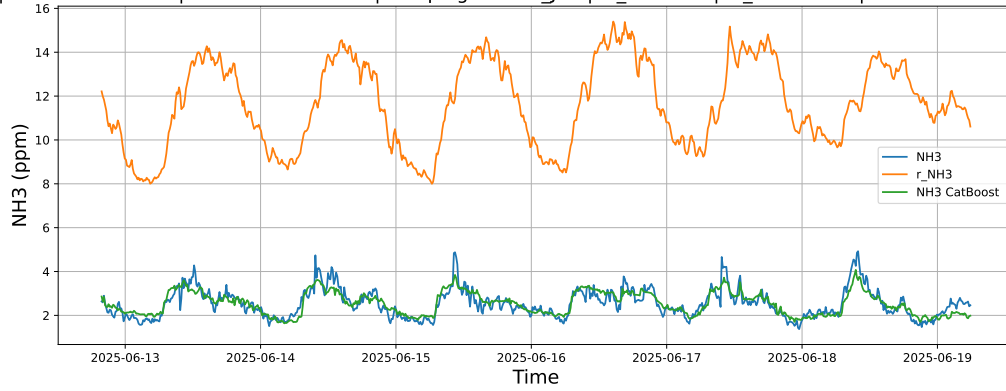
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

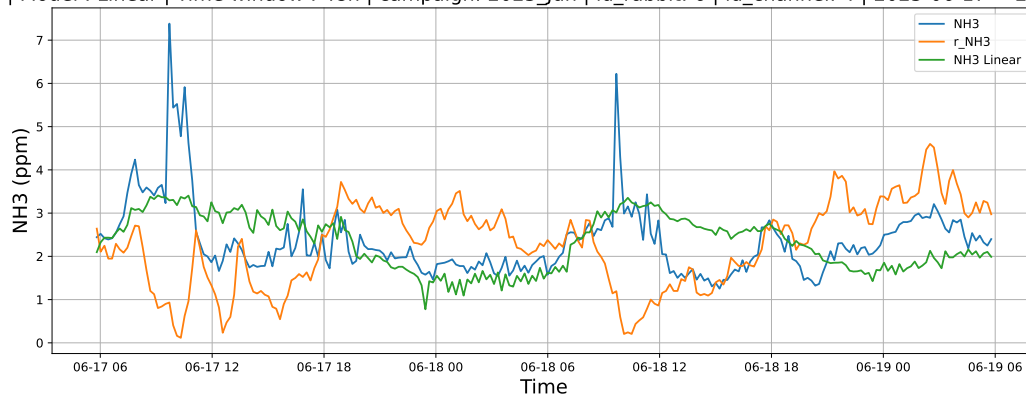


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

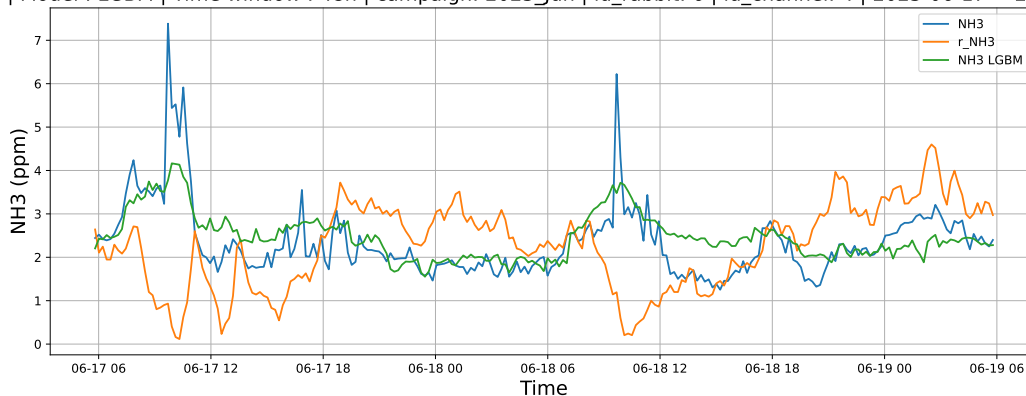


4.1.4.2 Time window : 48

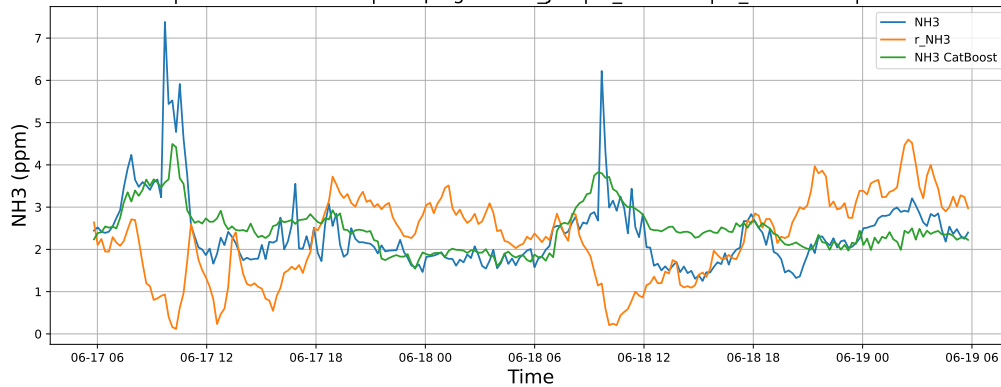
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

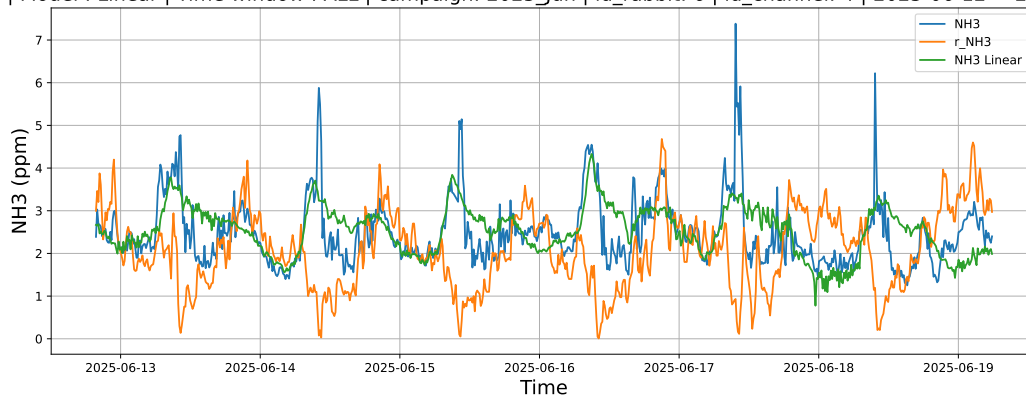


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

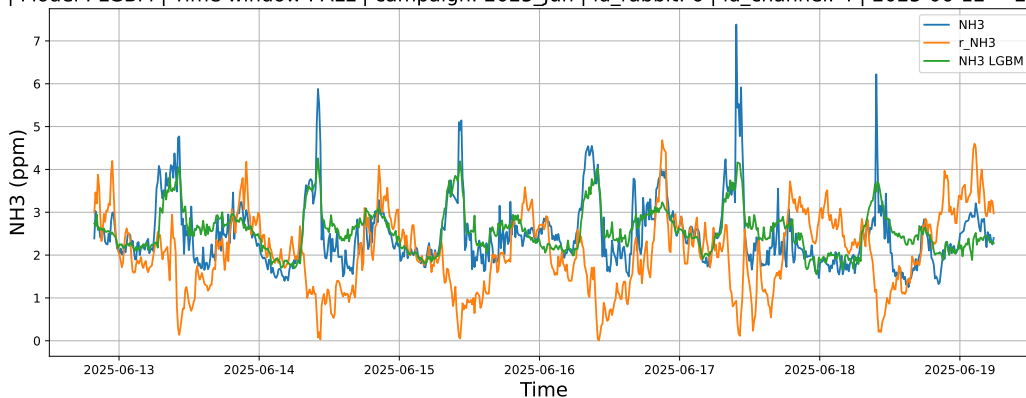


4.1.4.3 Time window : ALL

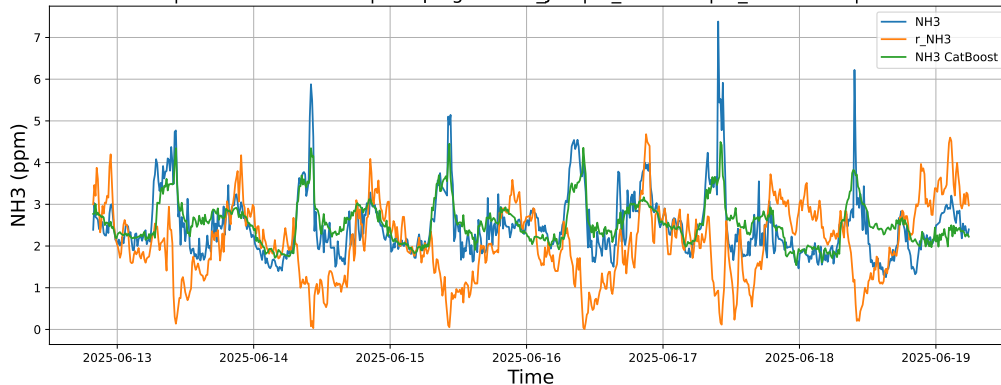
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



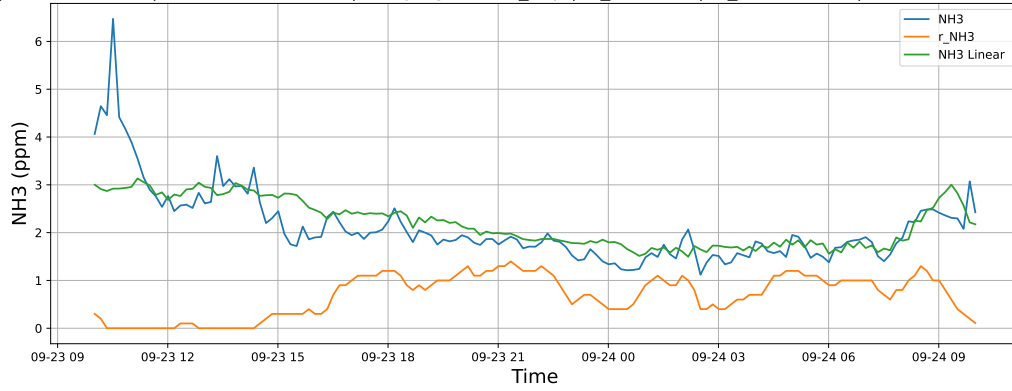
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



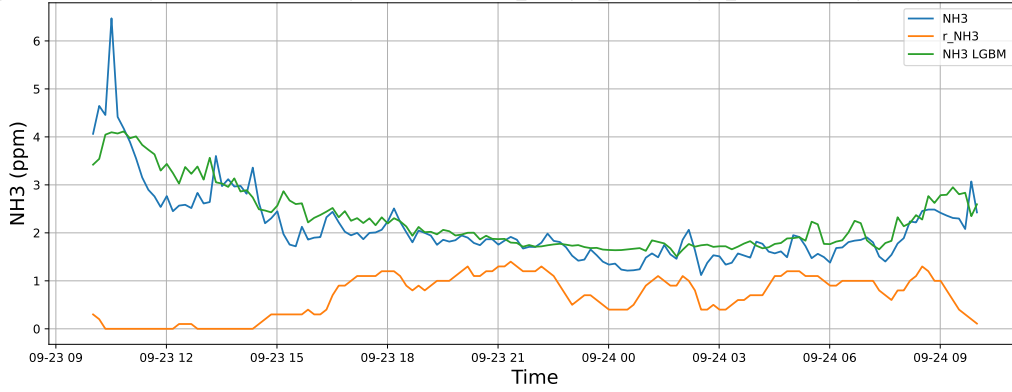
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

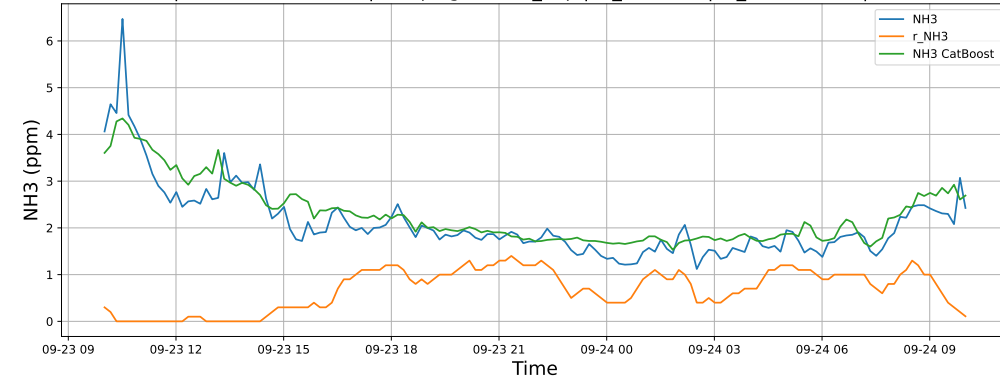
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

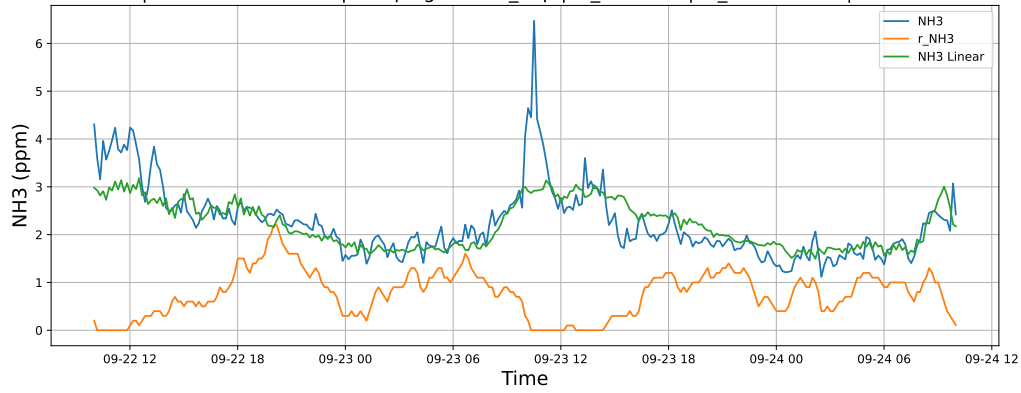


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

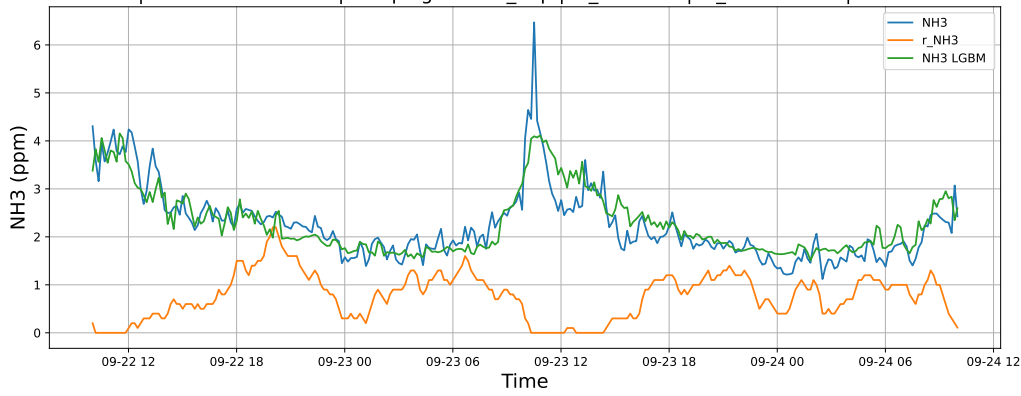


4.1.5.2 Time window : 48

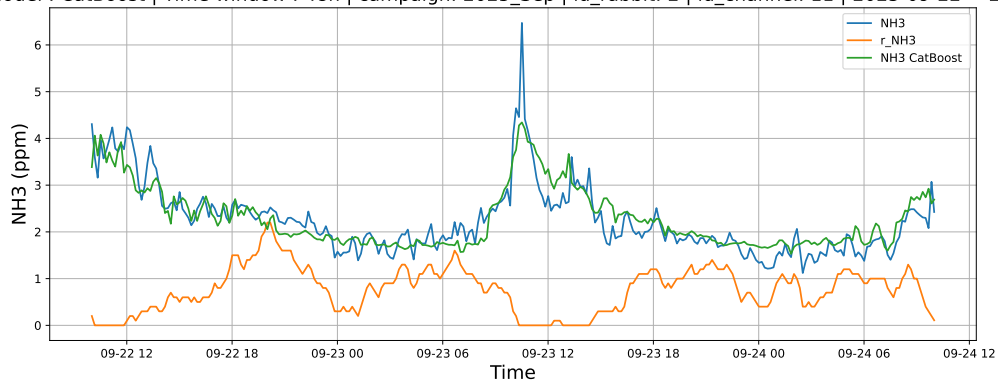
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

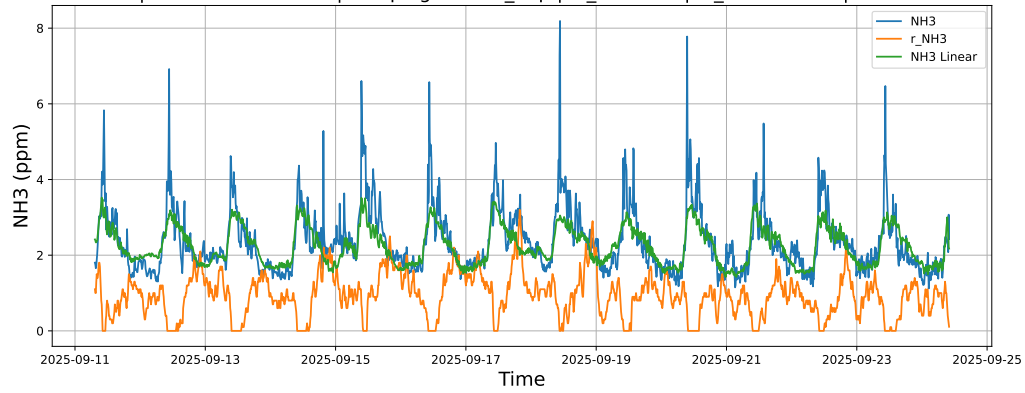


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

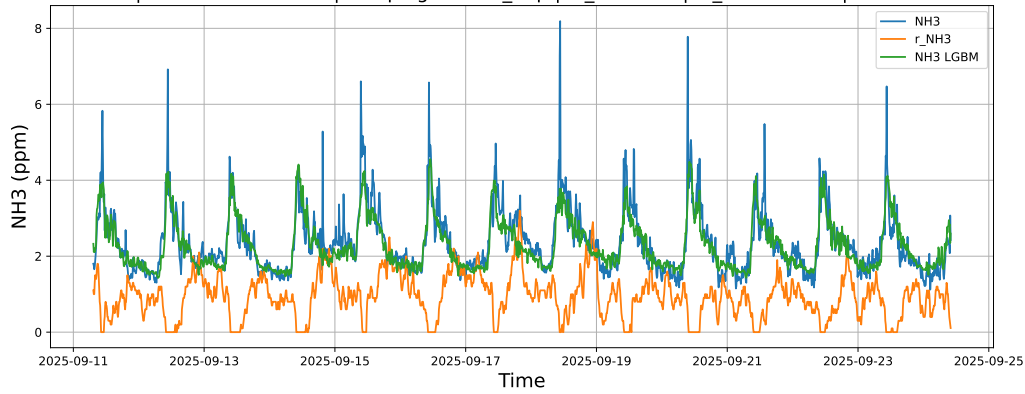


4.1.5.3 Time window : ALL

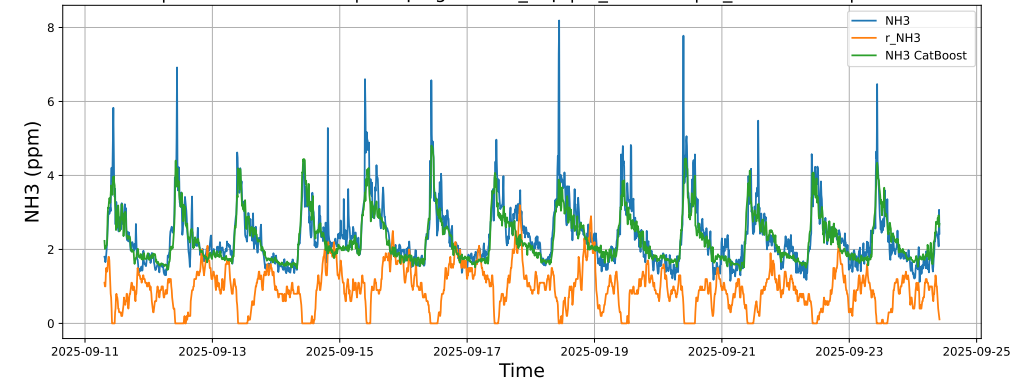
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



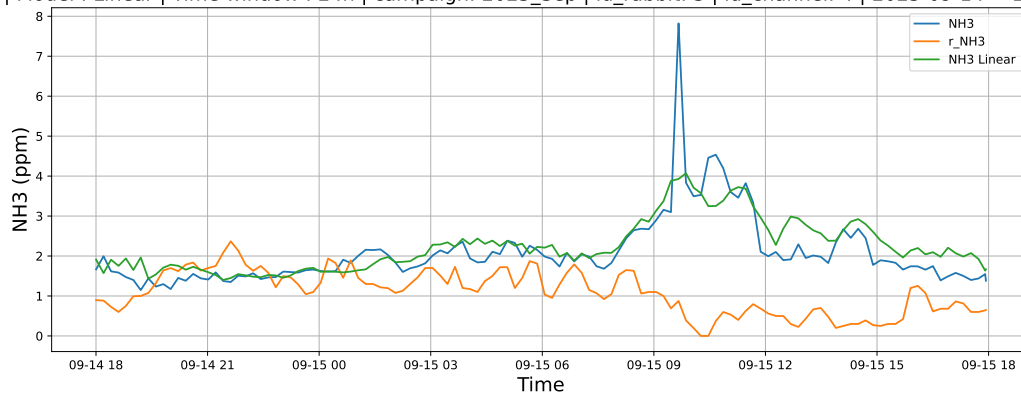
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



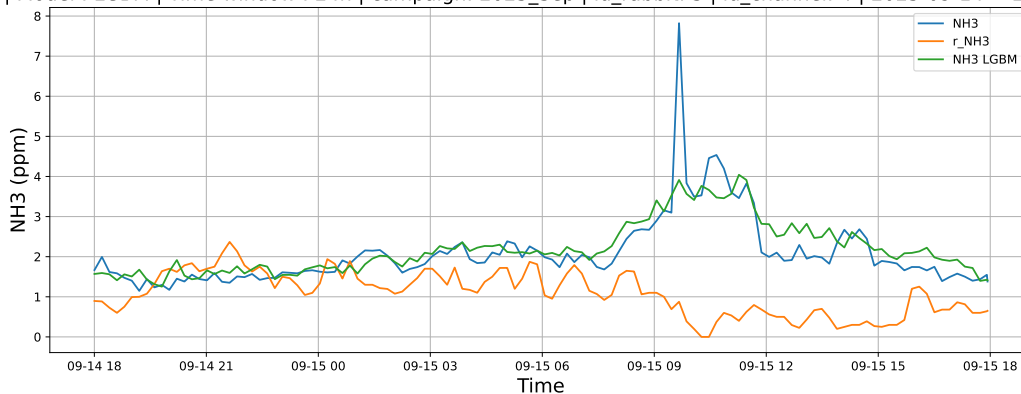
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

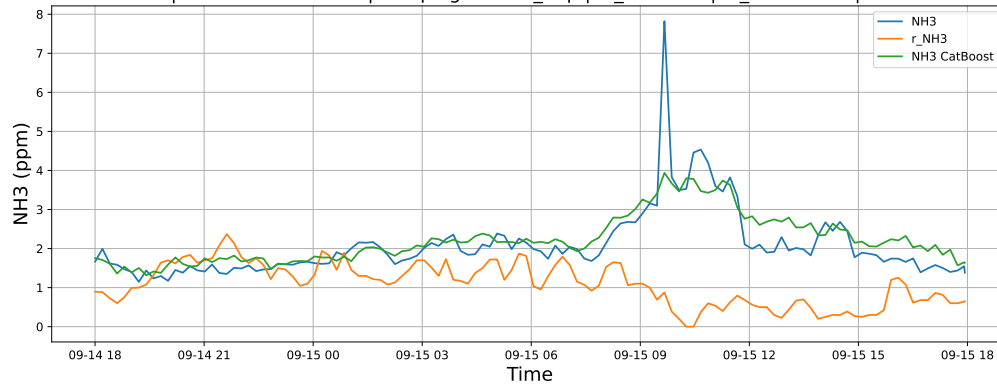
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

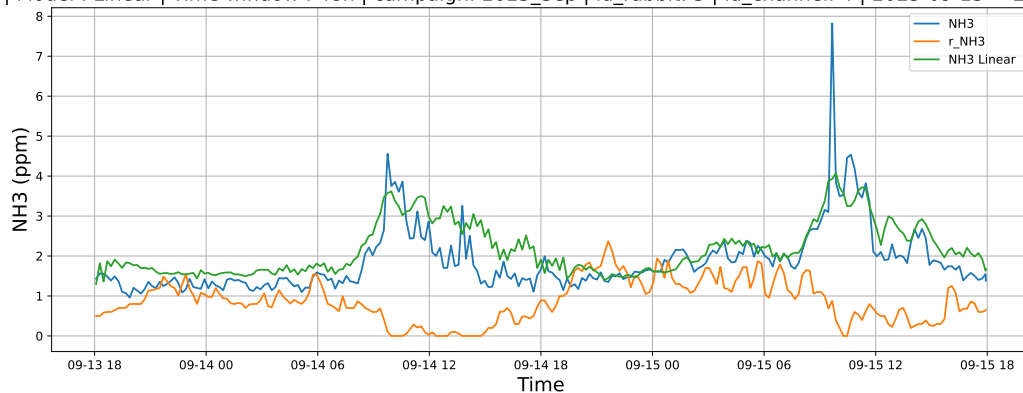


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

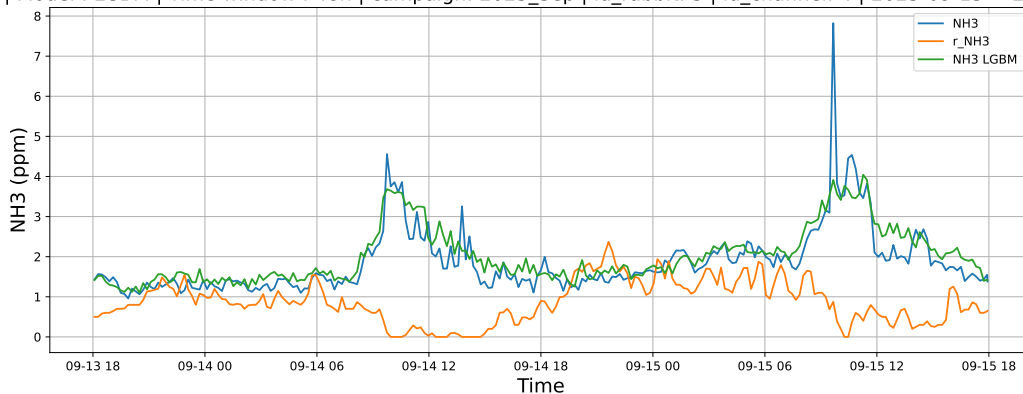


4.1.6.2 Time window : 48

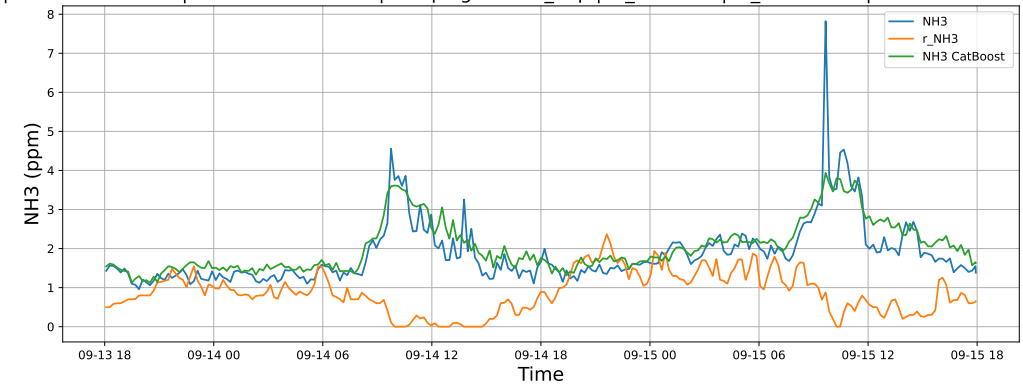
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

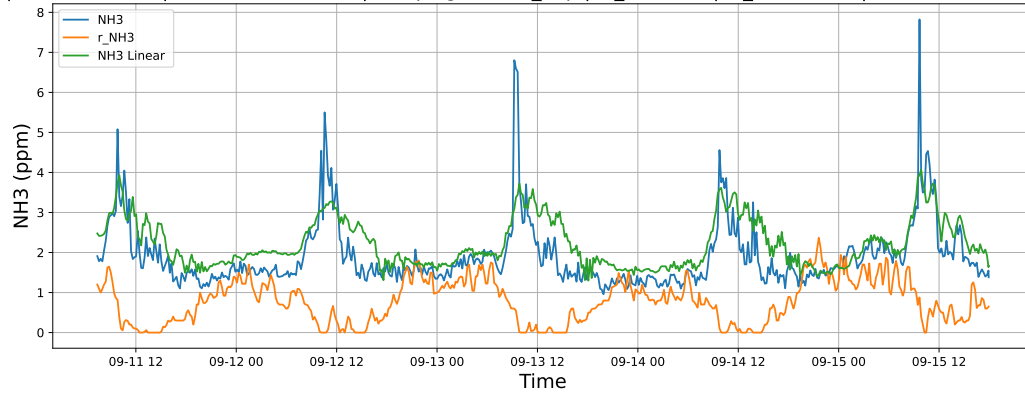


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

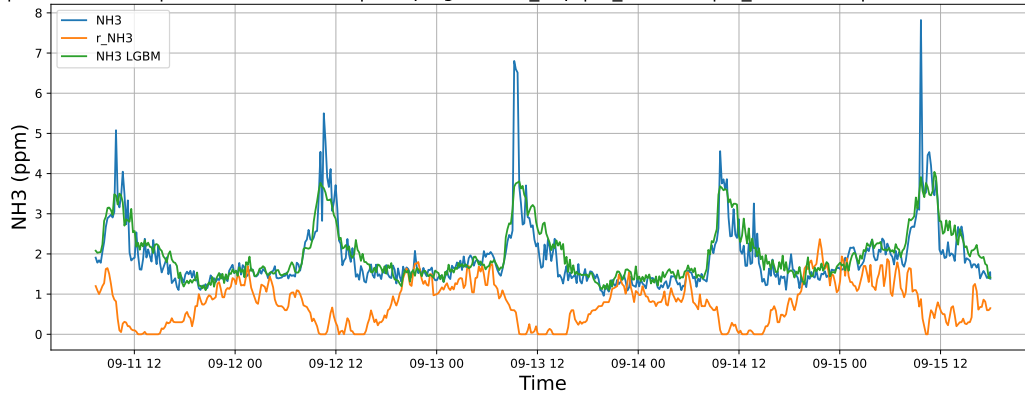


4.1.6.3 Time window : ALL

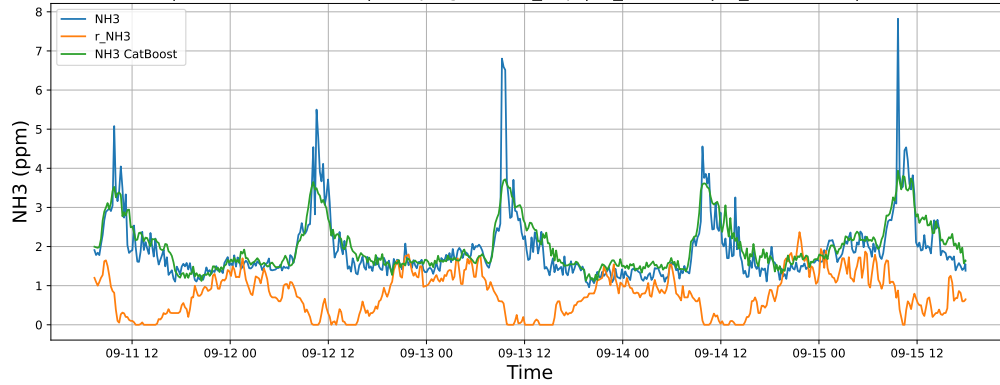
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



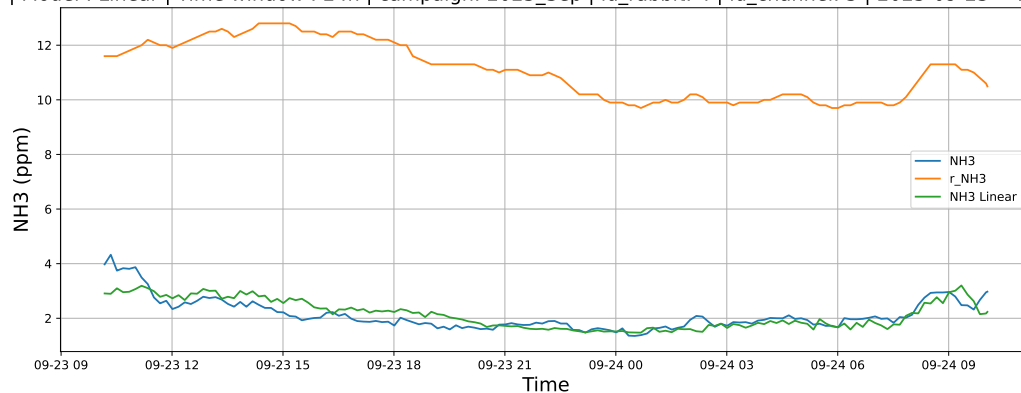
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



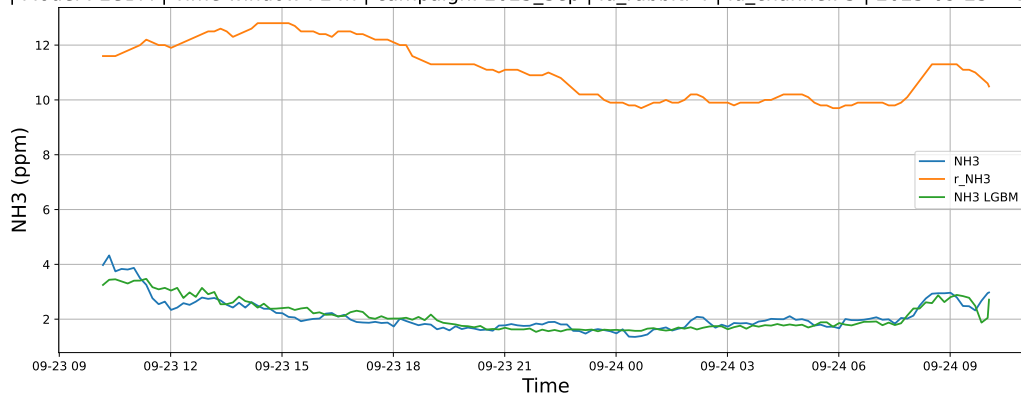
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

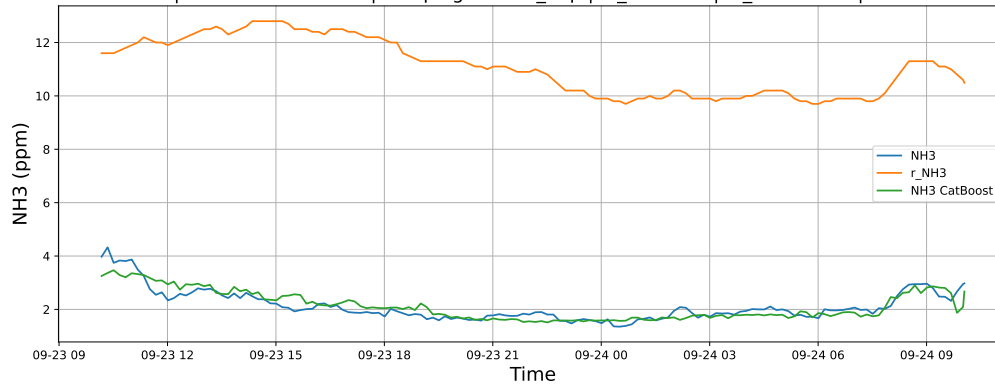
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

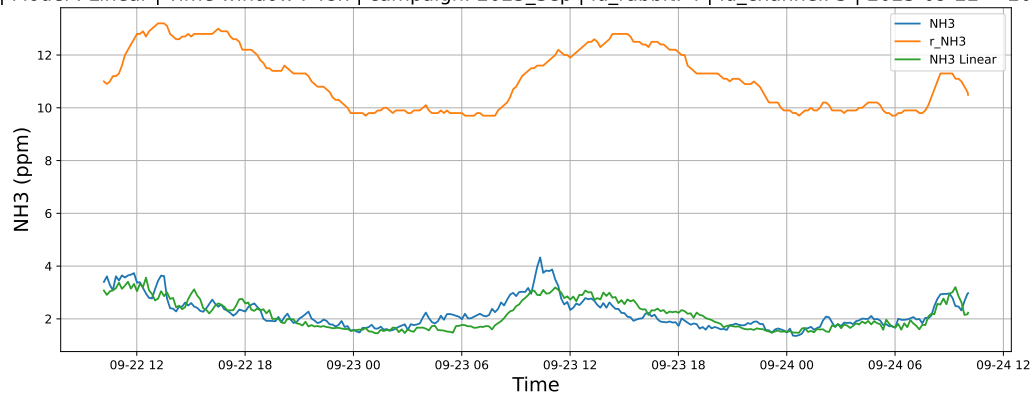


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

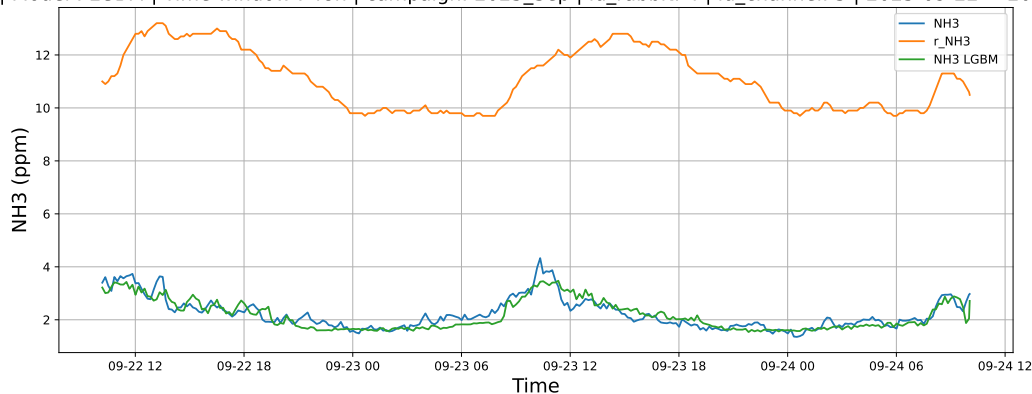


4.1.7.2 Time window : 48

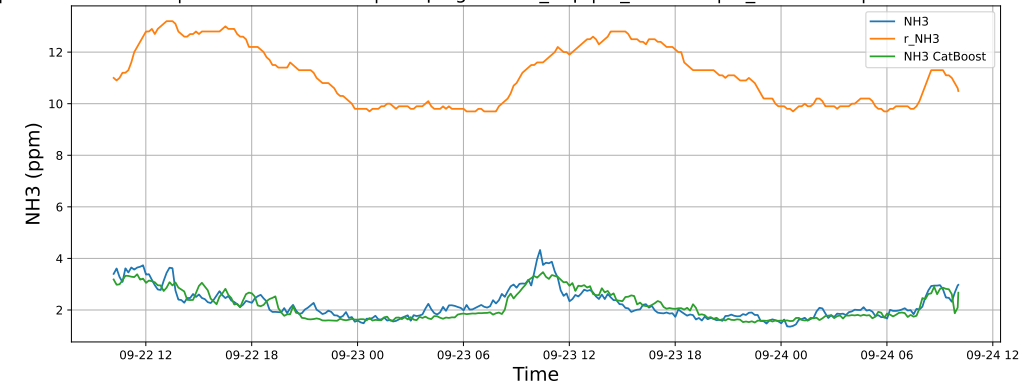
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

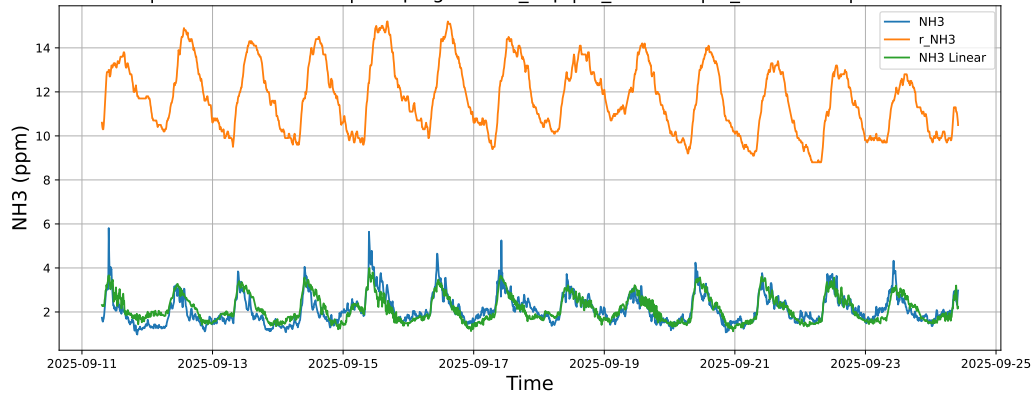


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

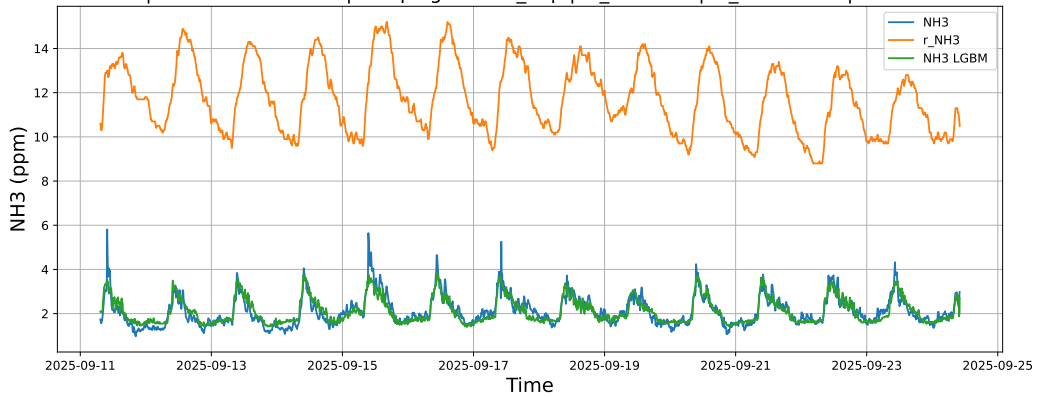


4.1.7.3 Time window : ALL

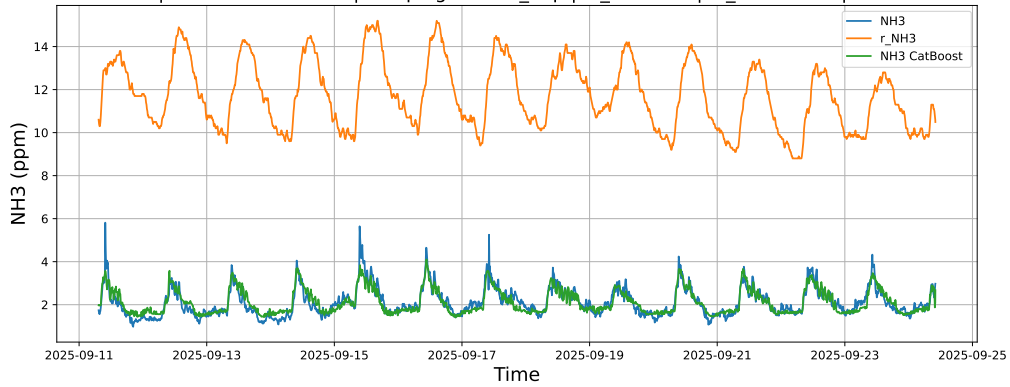
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



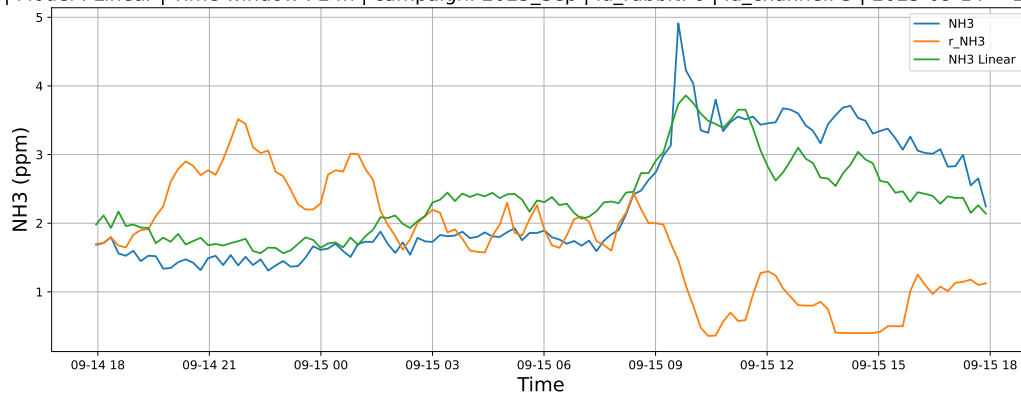
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



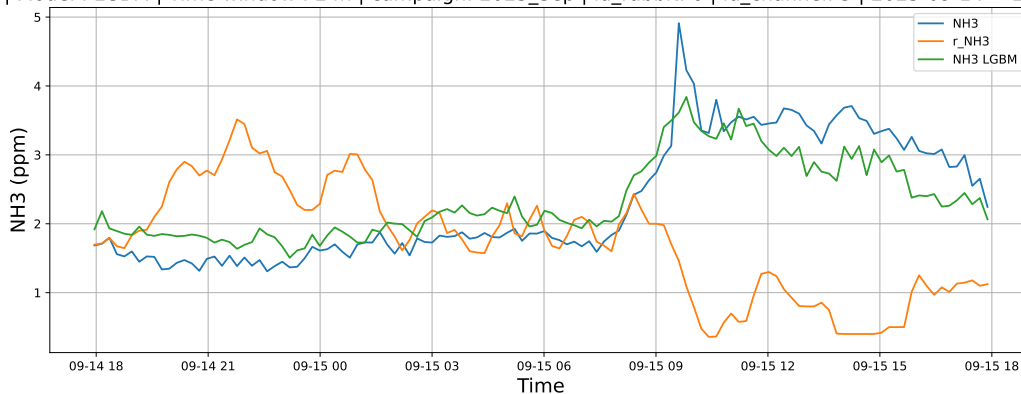
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

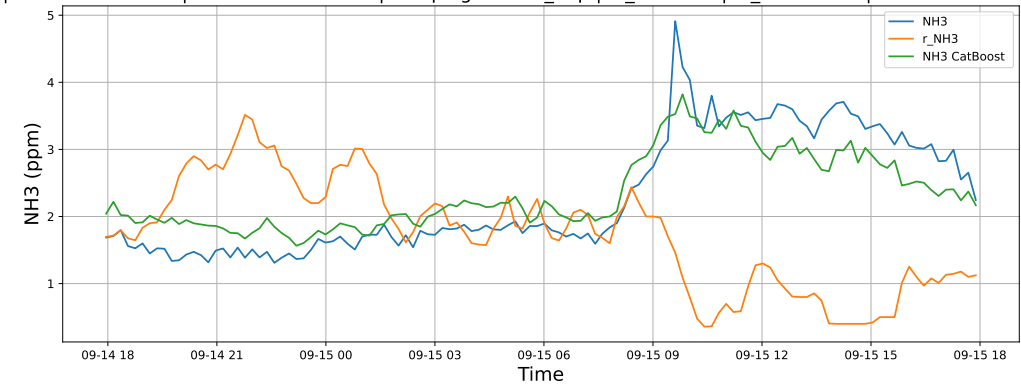
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

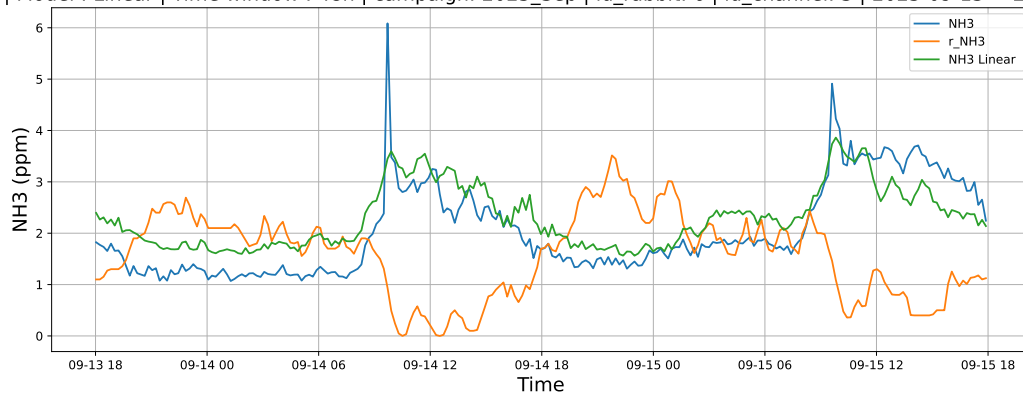


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

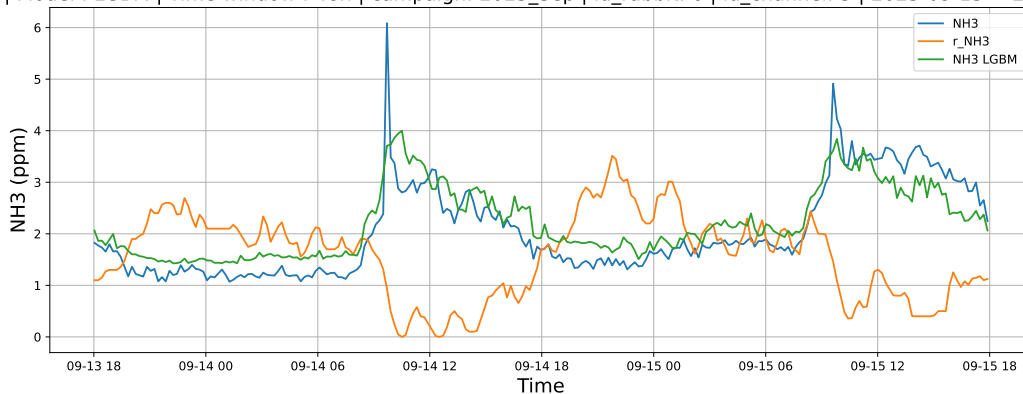


4.1.8.2 Time window : 48

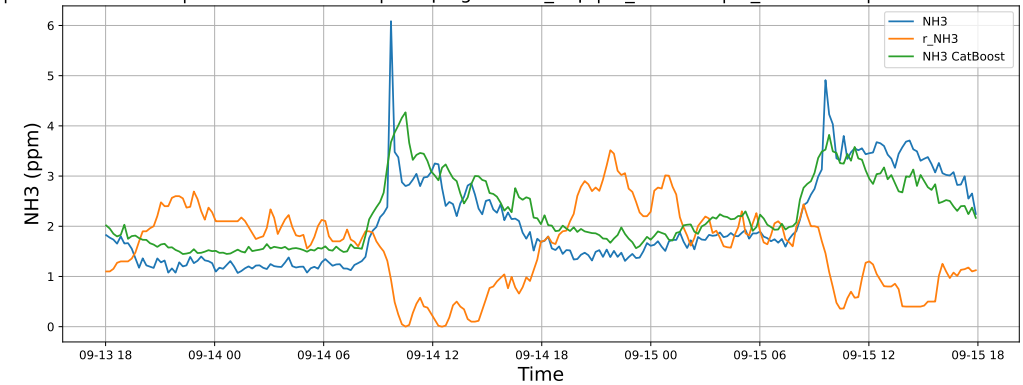
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

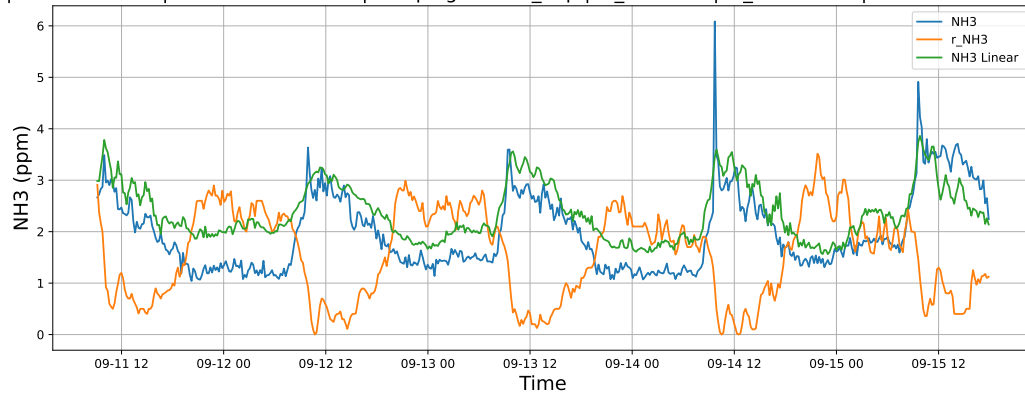


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

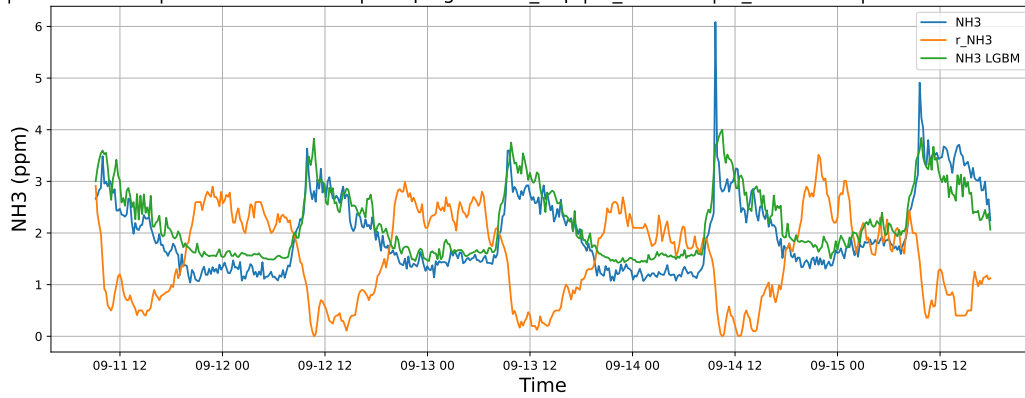


4.1.8.3 Time window : ALL

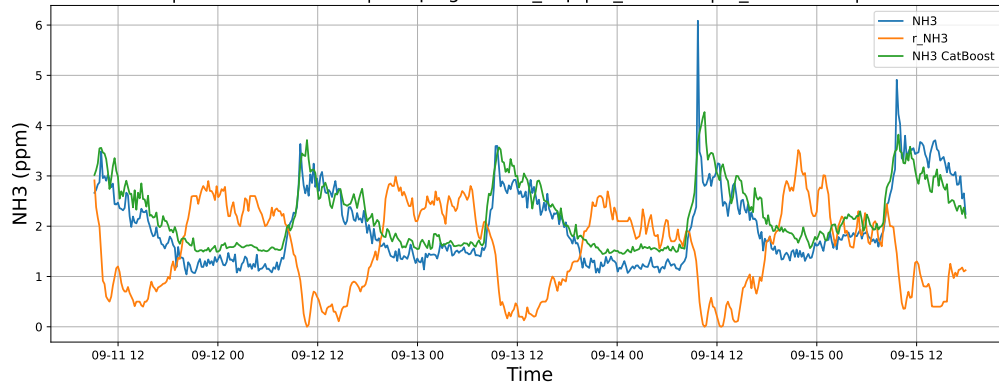
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

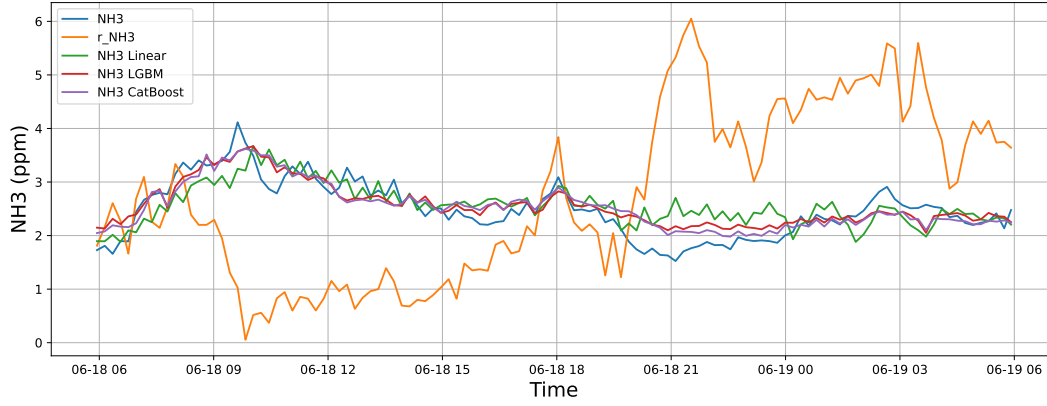


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

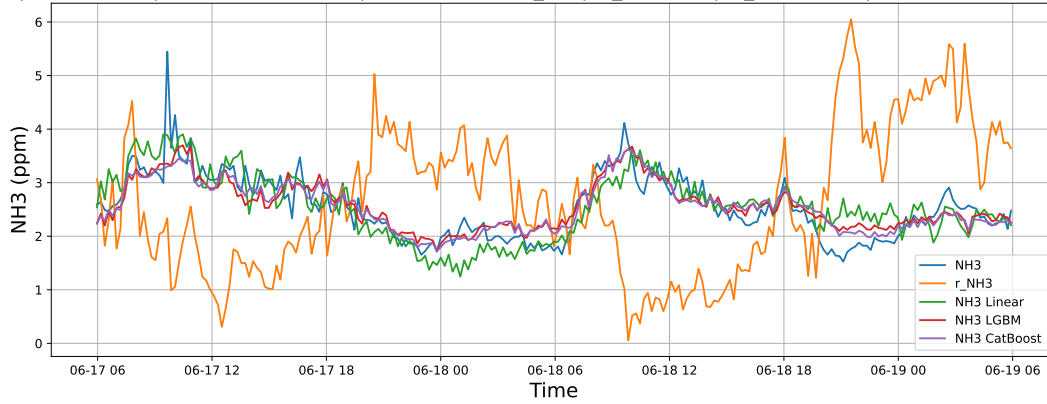
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



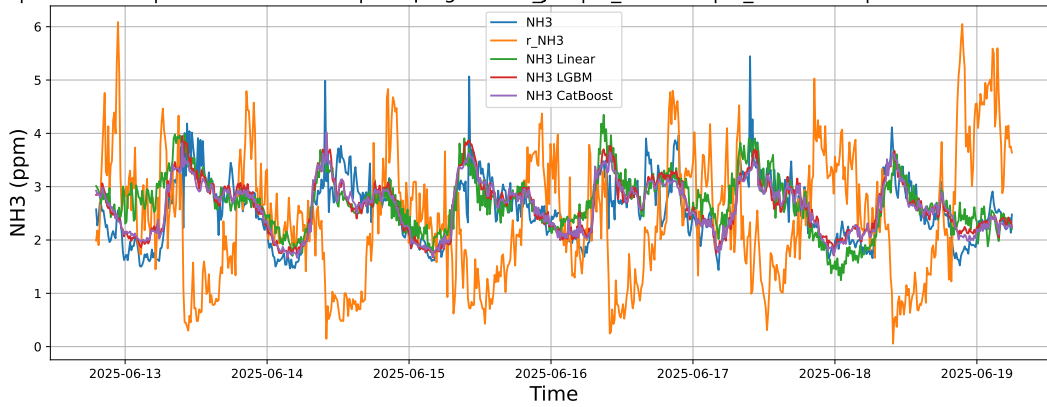
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

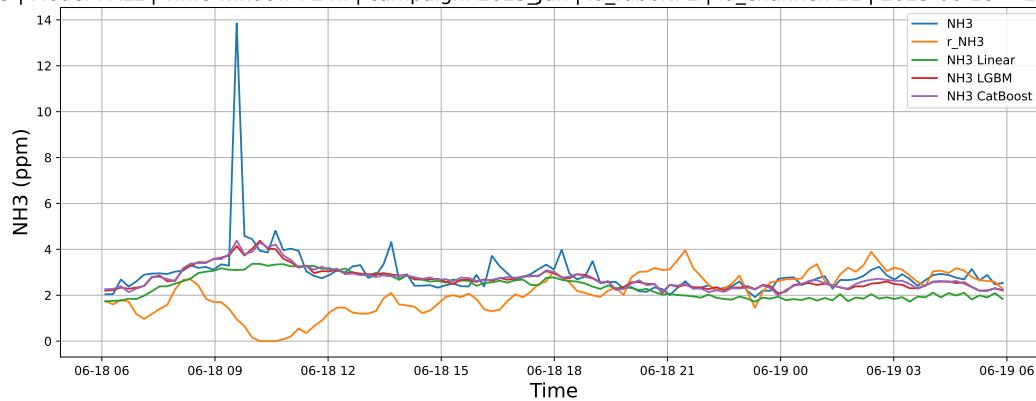
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

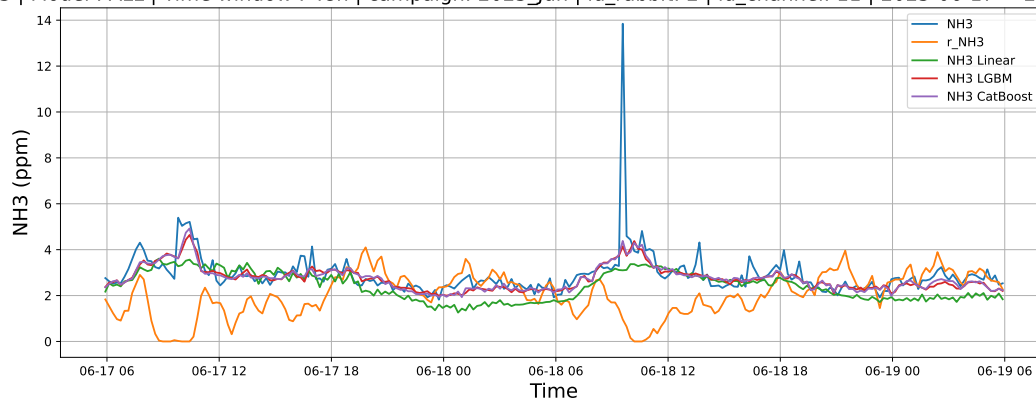
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



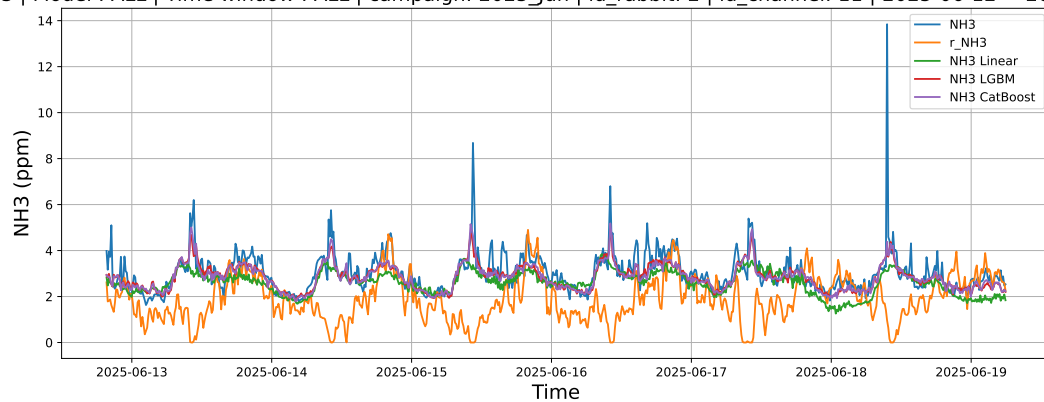
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

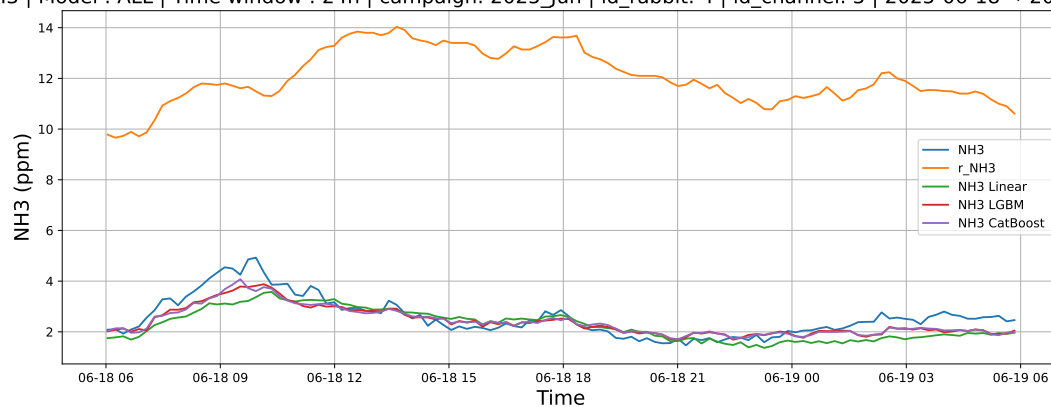
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

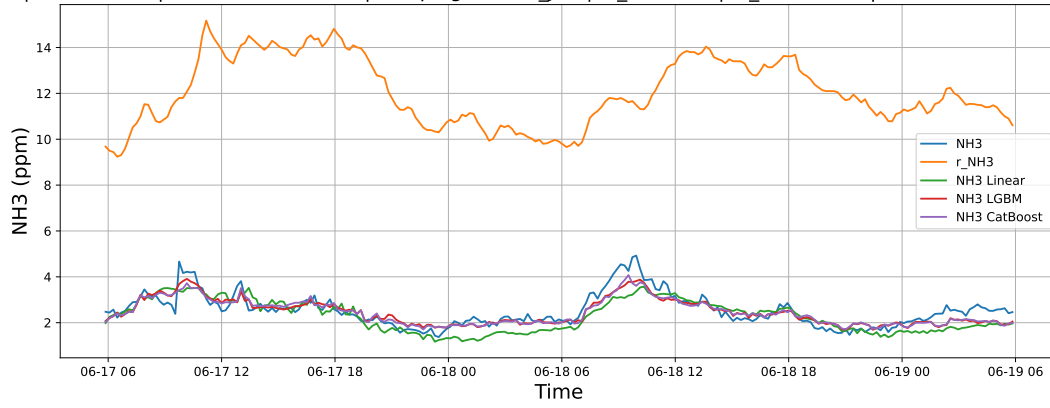
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



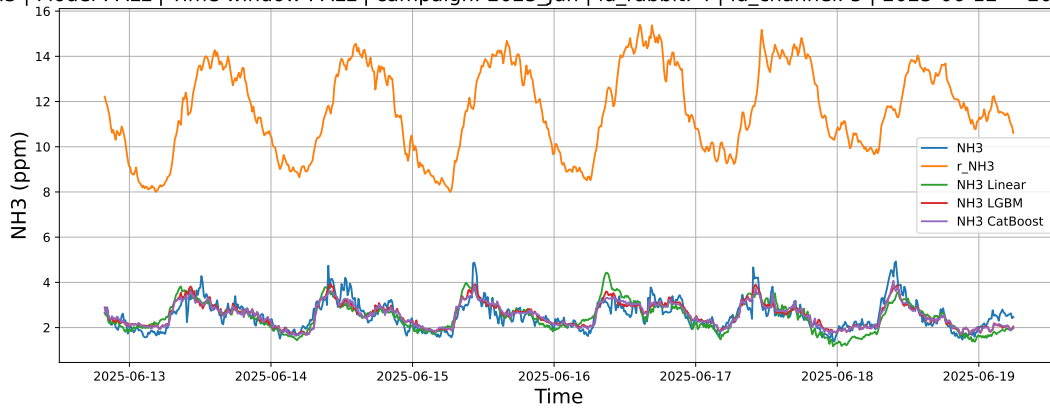
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

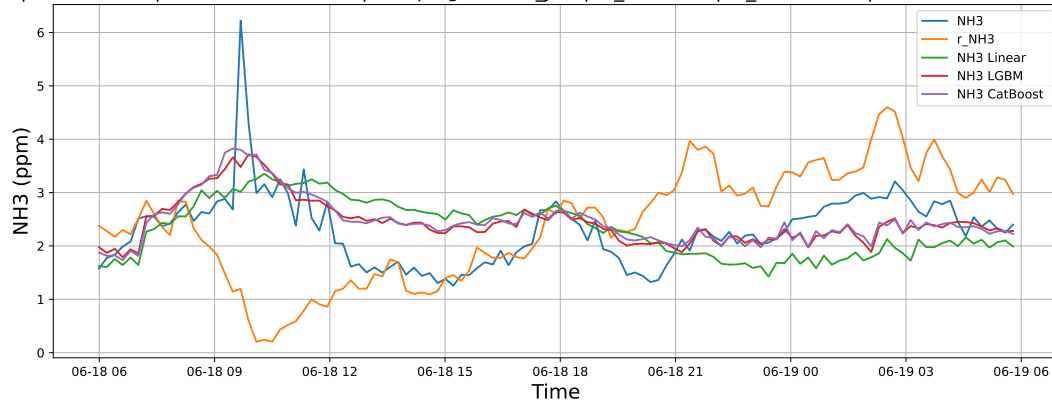
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

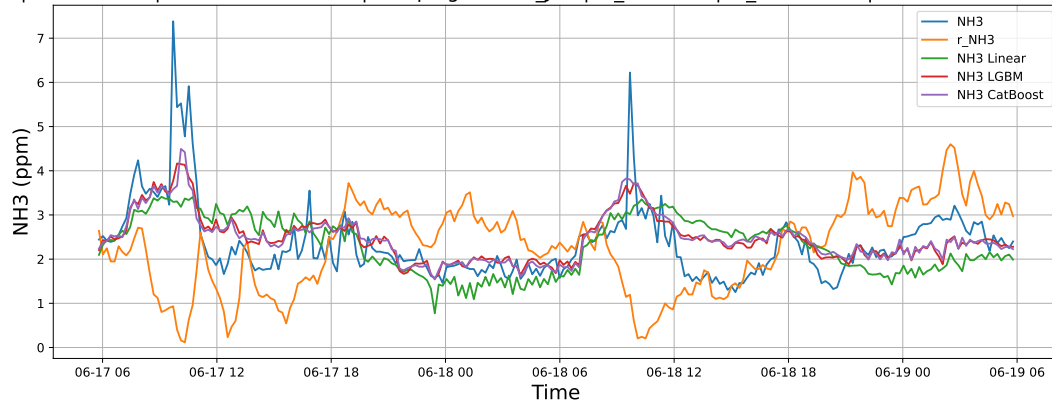
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



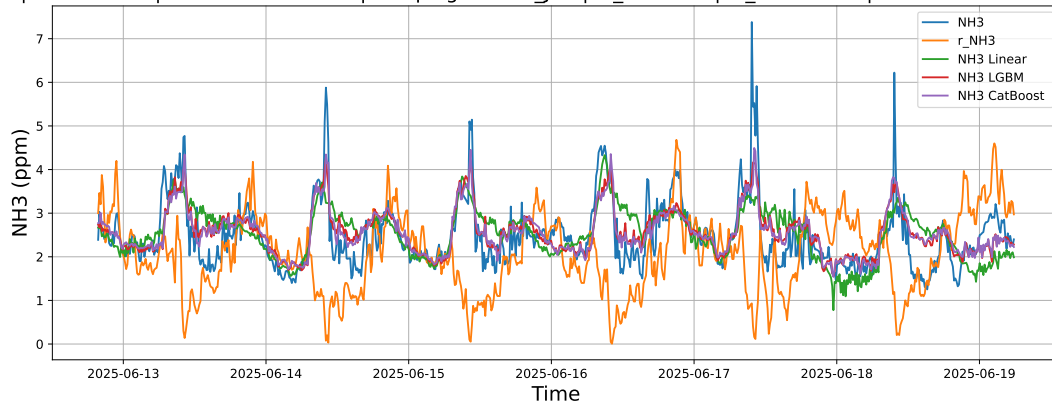
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

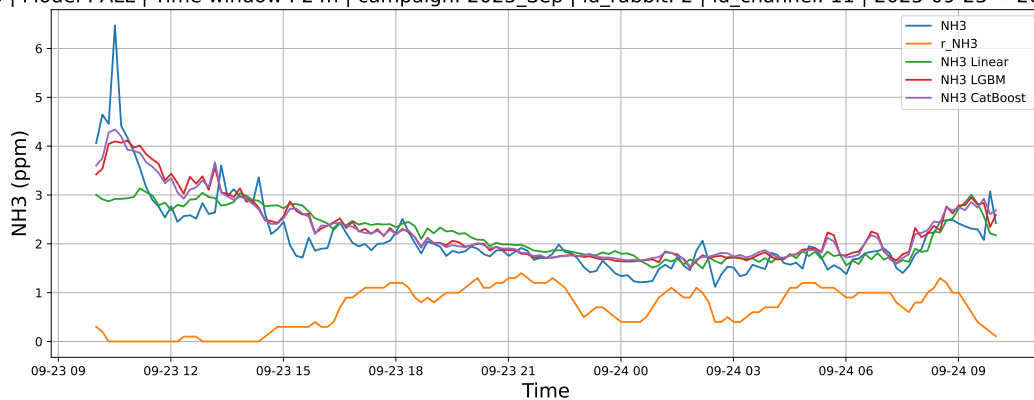
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

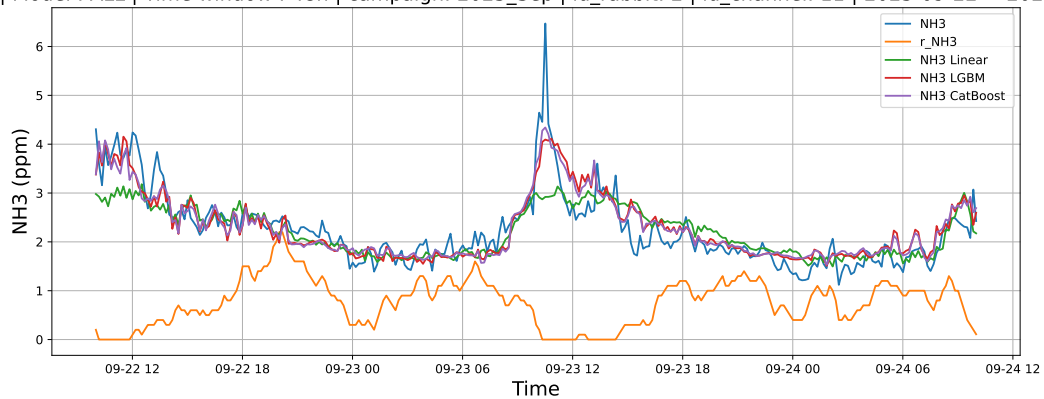
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



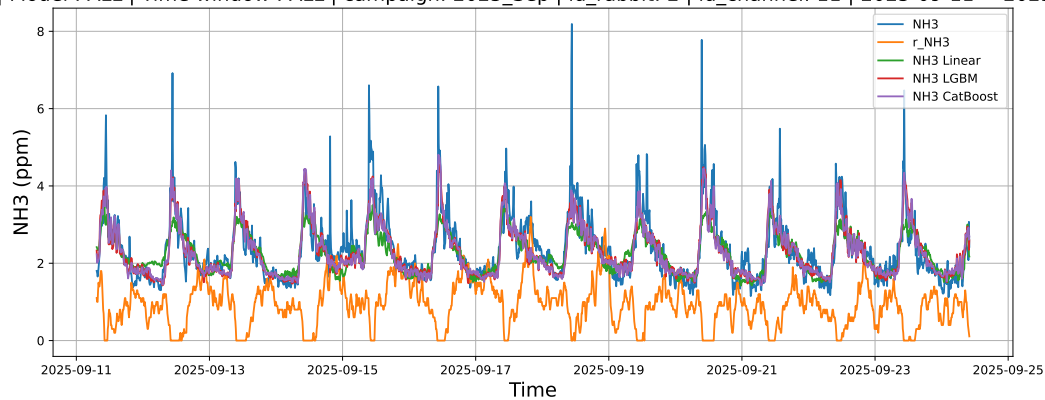
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

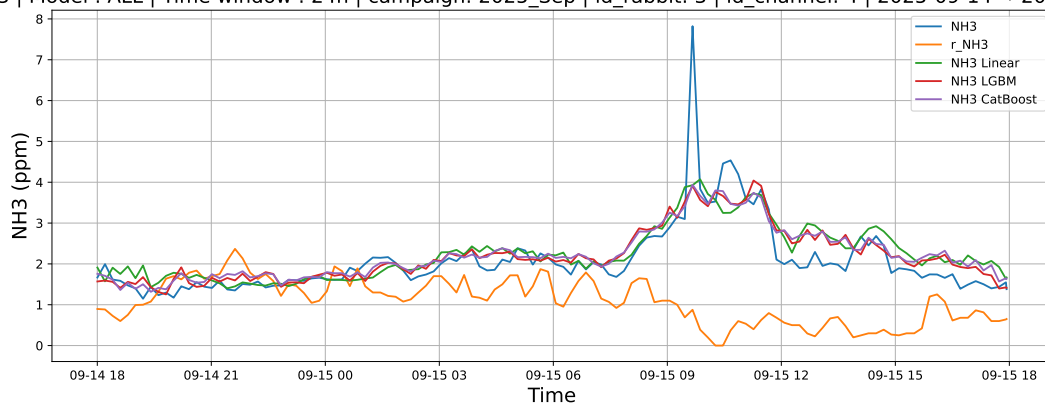
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

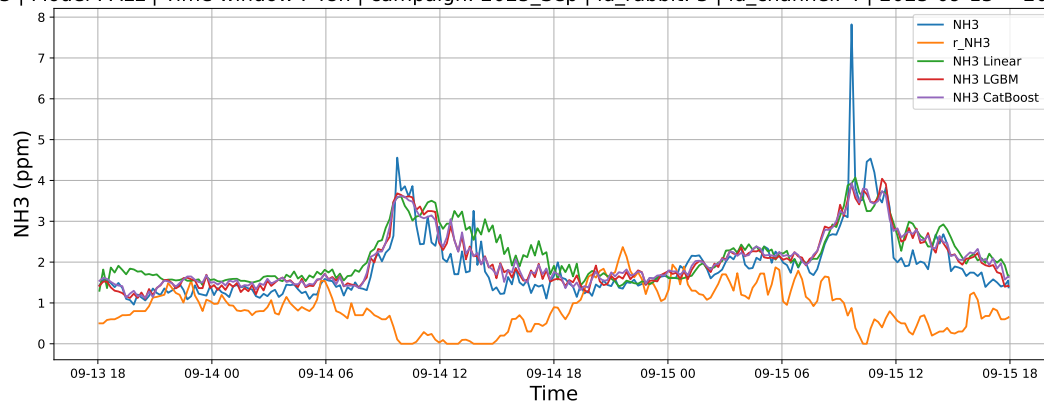
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



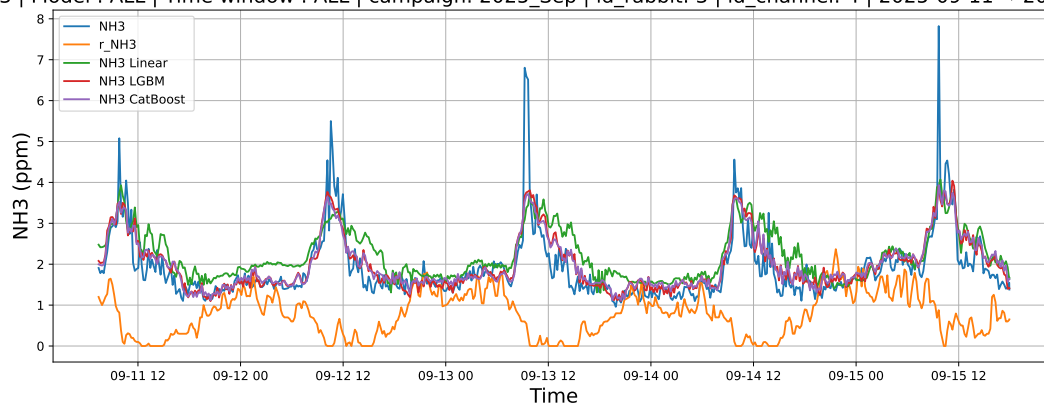
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

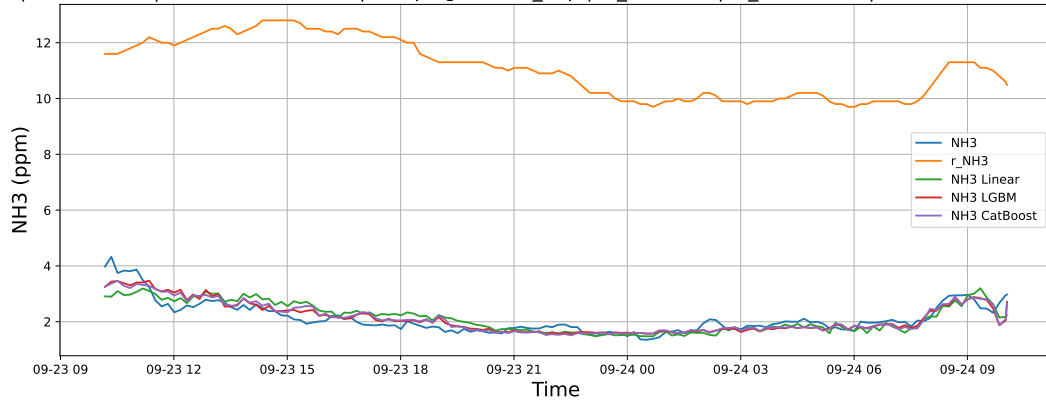
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

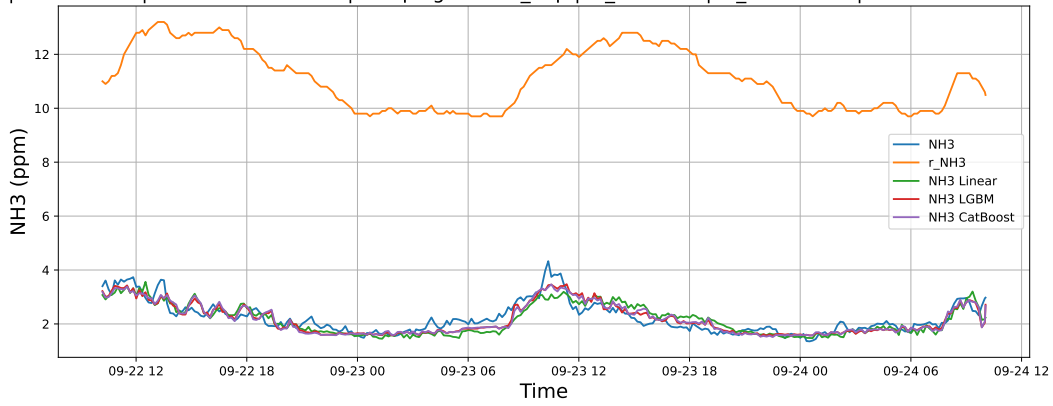
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



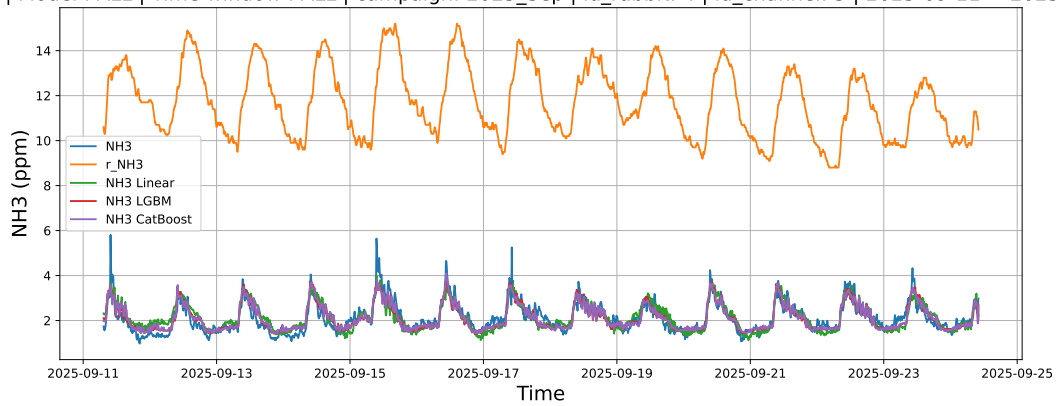
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

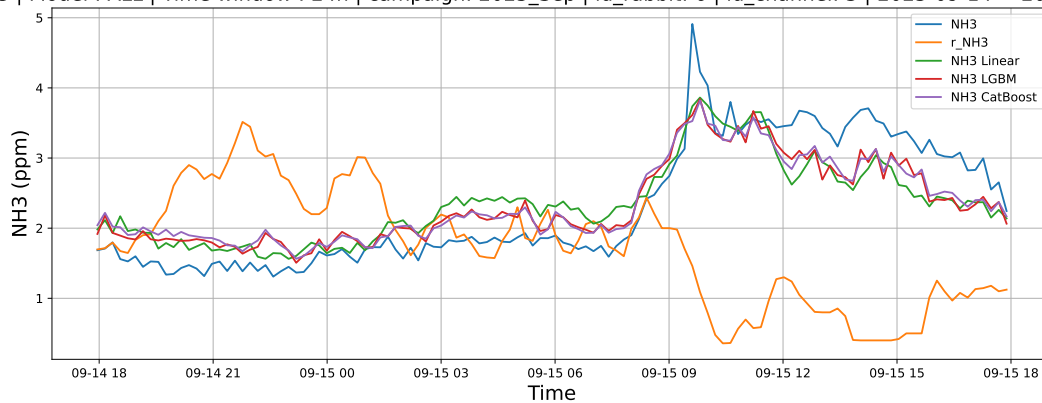
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

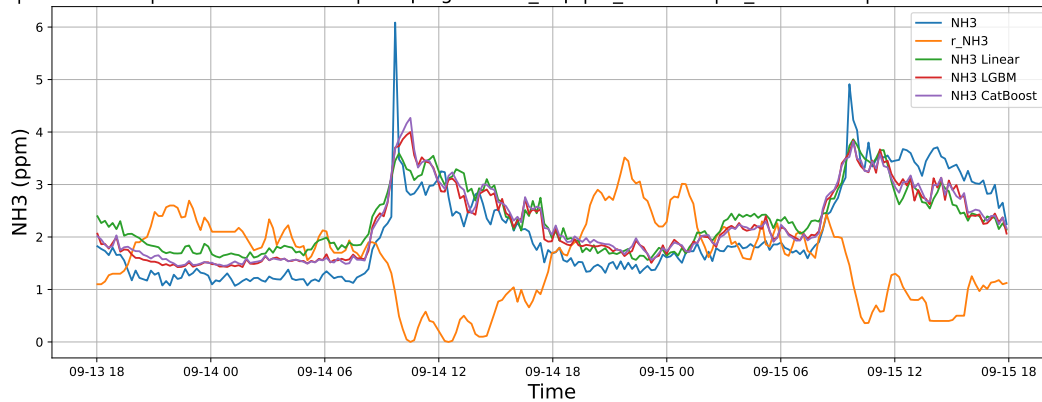
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

